

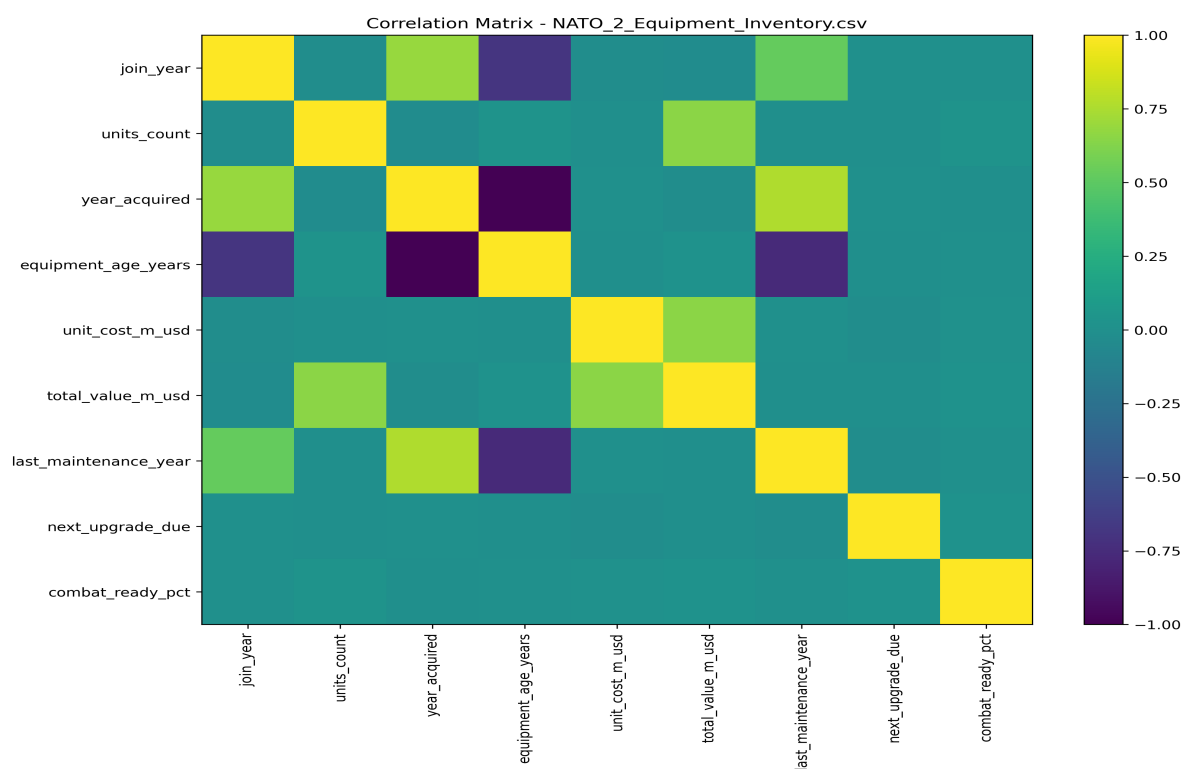
NATO Alliance Complete Dataset 2024

Advanced Data Science, Time-Series and Machine Learning Analysis

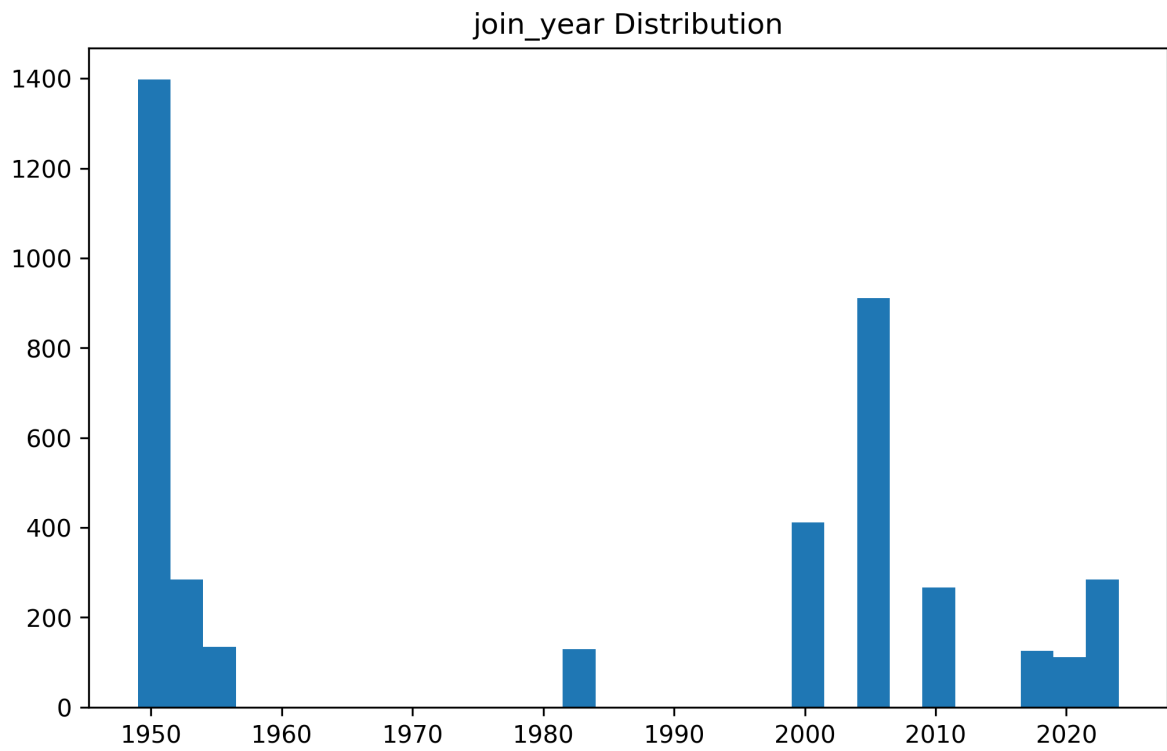
This report presents statistically meaningful exploratory data analysis, machine learning, clustering, anomaly detection, NATO-specific insights, and temporal trend analysis for NATO datasets.

Dataset Analysis: NATO_2_Equipment_Inventory.csv

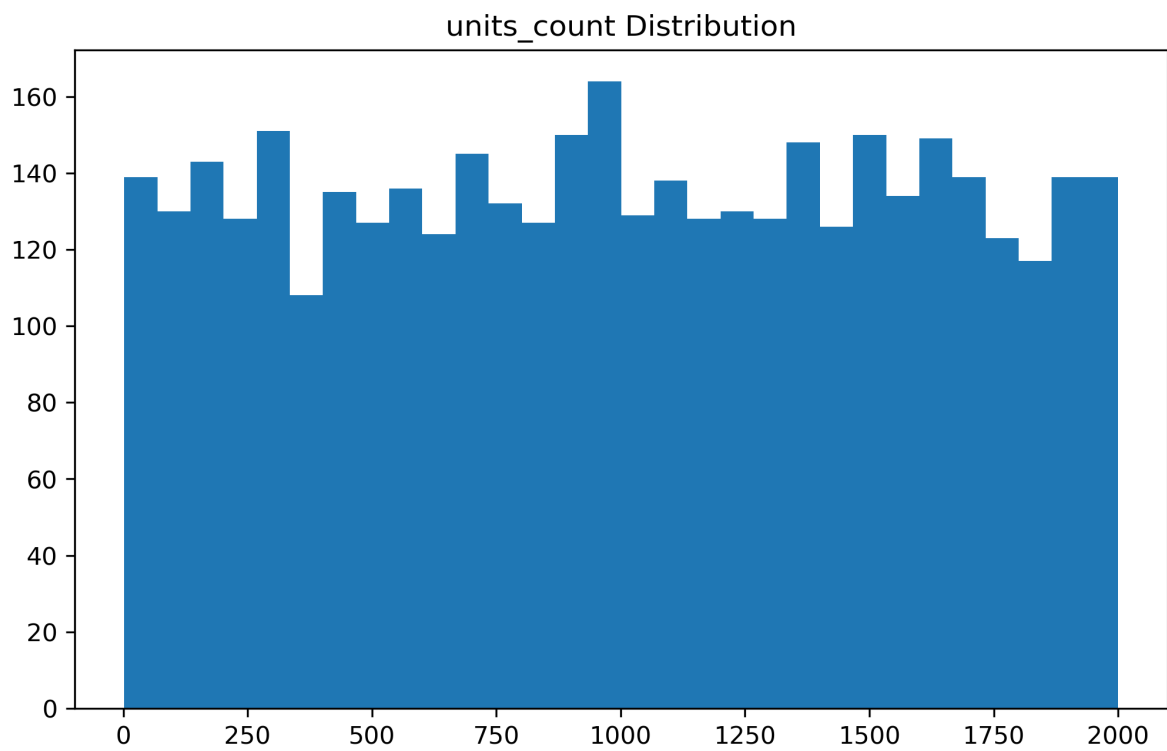
Rows: 4056
Columns: 26
Target Variable: combat_ready_pct
Numerical Features: 9
Categorical Features: 15
Generated Charts: 17
Optimal Clusters: 2
Silhouette Score: 0.141
Random Forest RMSE: 17.365
Gradient Boosting RMSE: 17.241



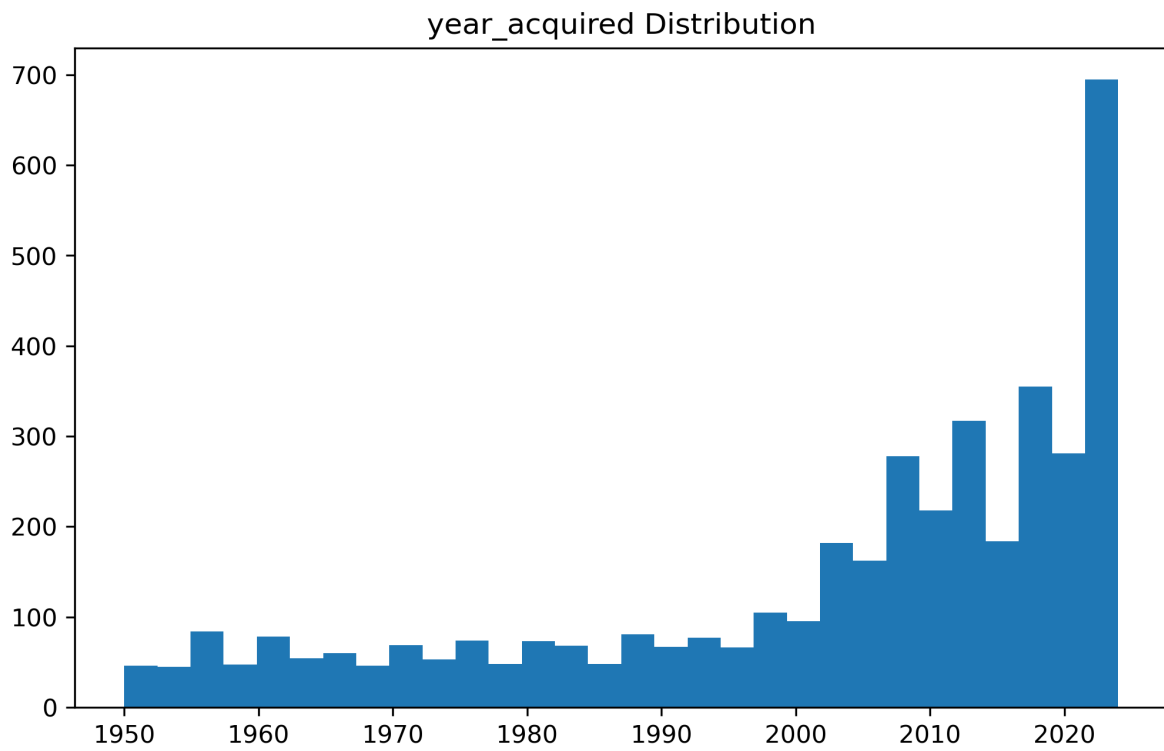
Correlation matrix of valid numerical variables only.



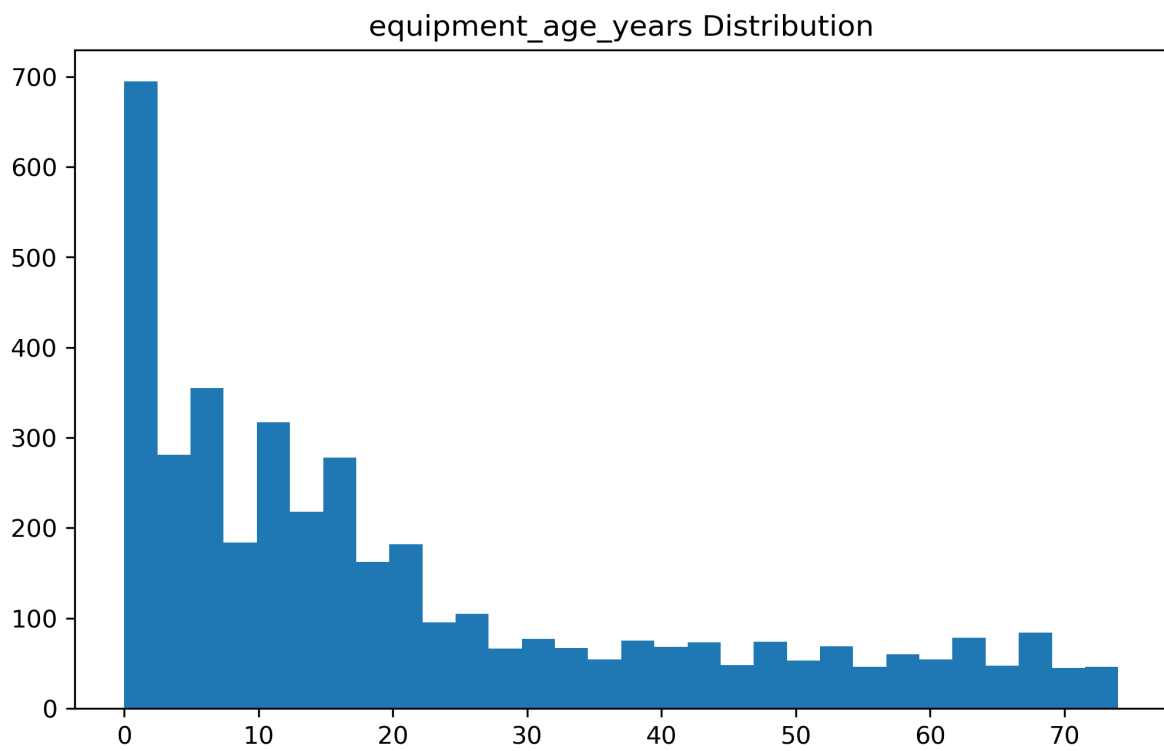
Distribution of join_year.



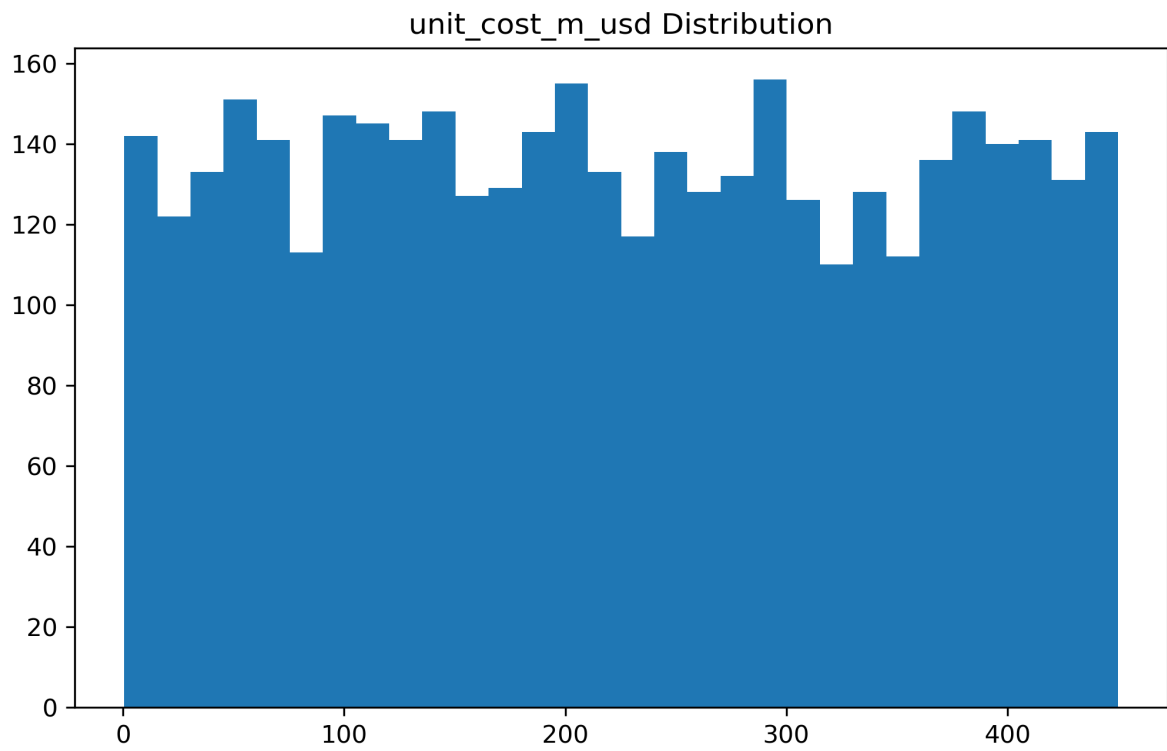
Distribution of units_count.



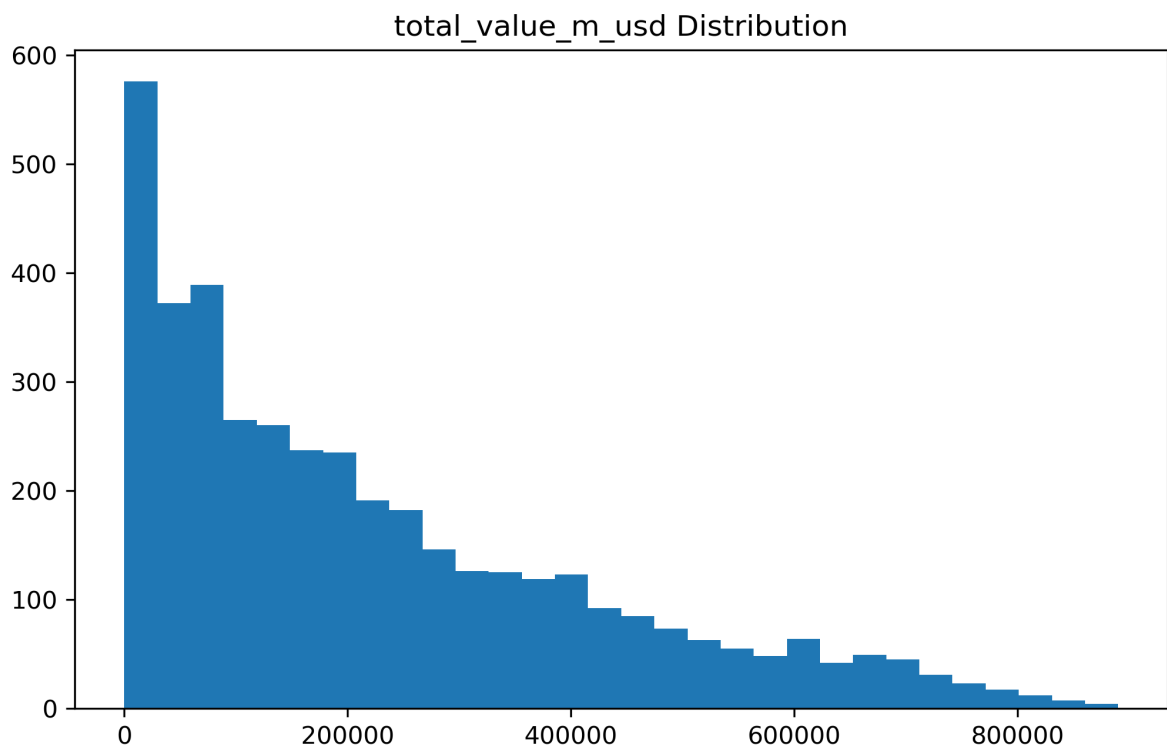
Distribution of year_acquired.



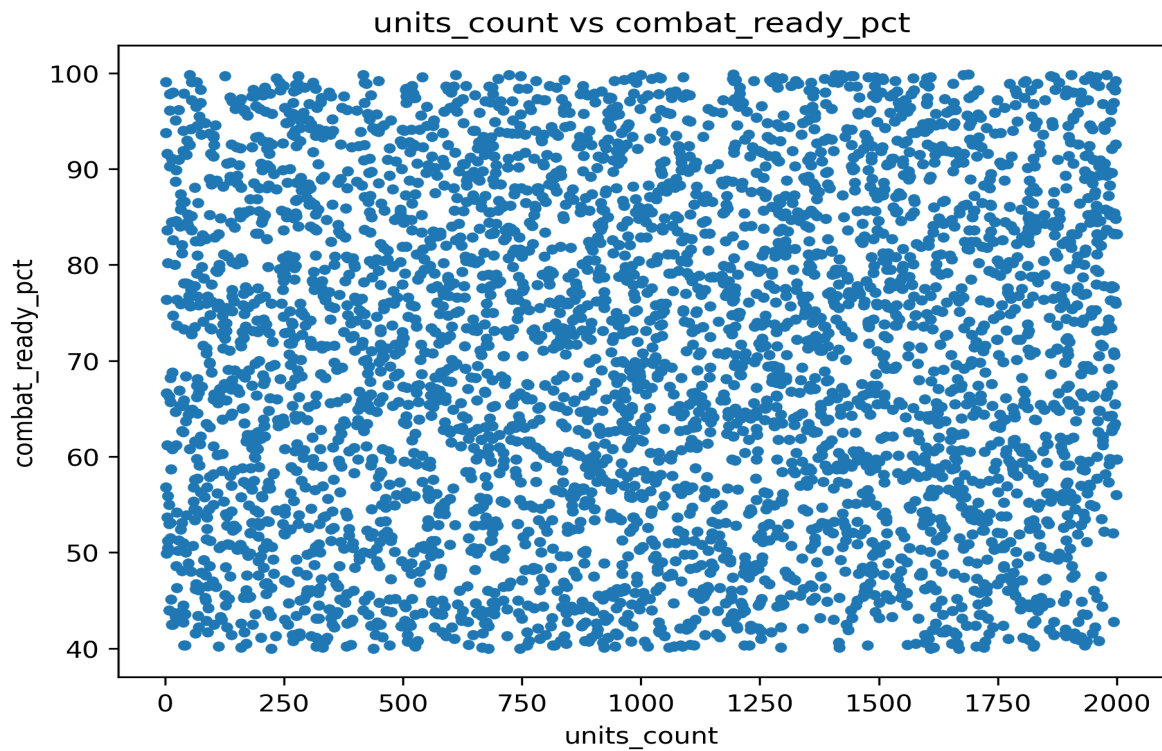
Distribution of equipment_age_years.



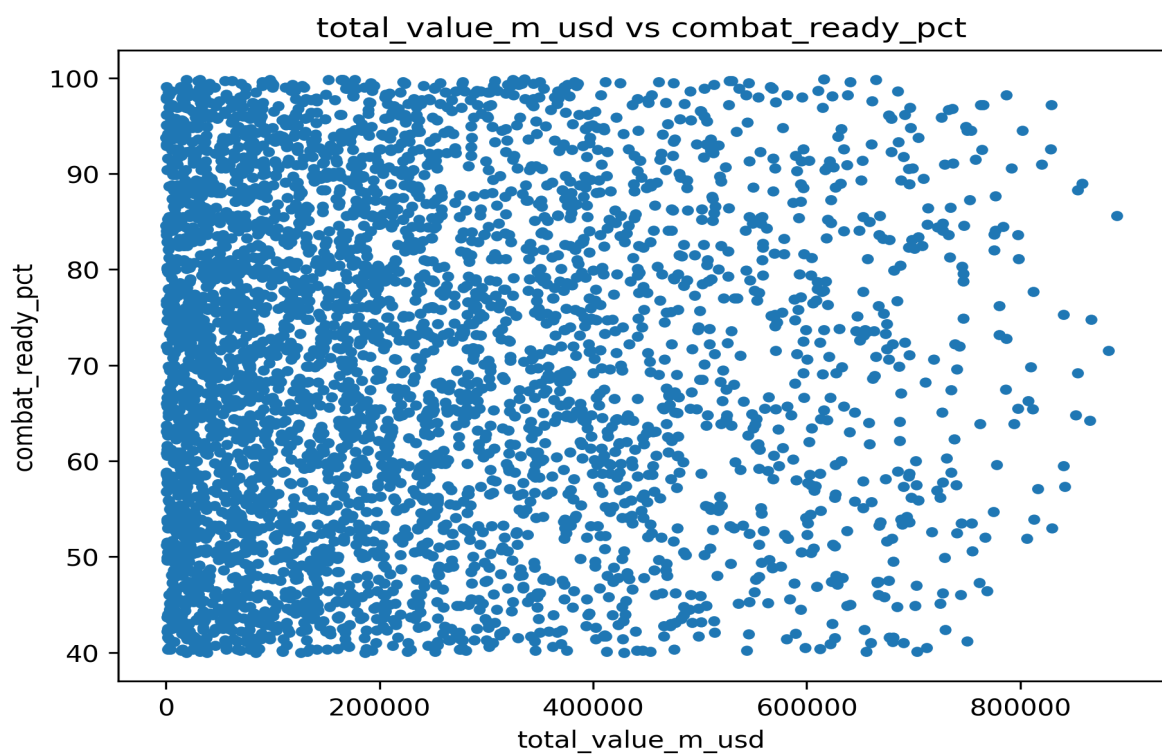
Distribution of unit_cost_m_usd.



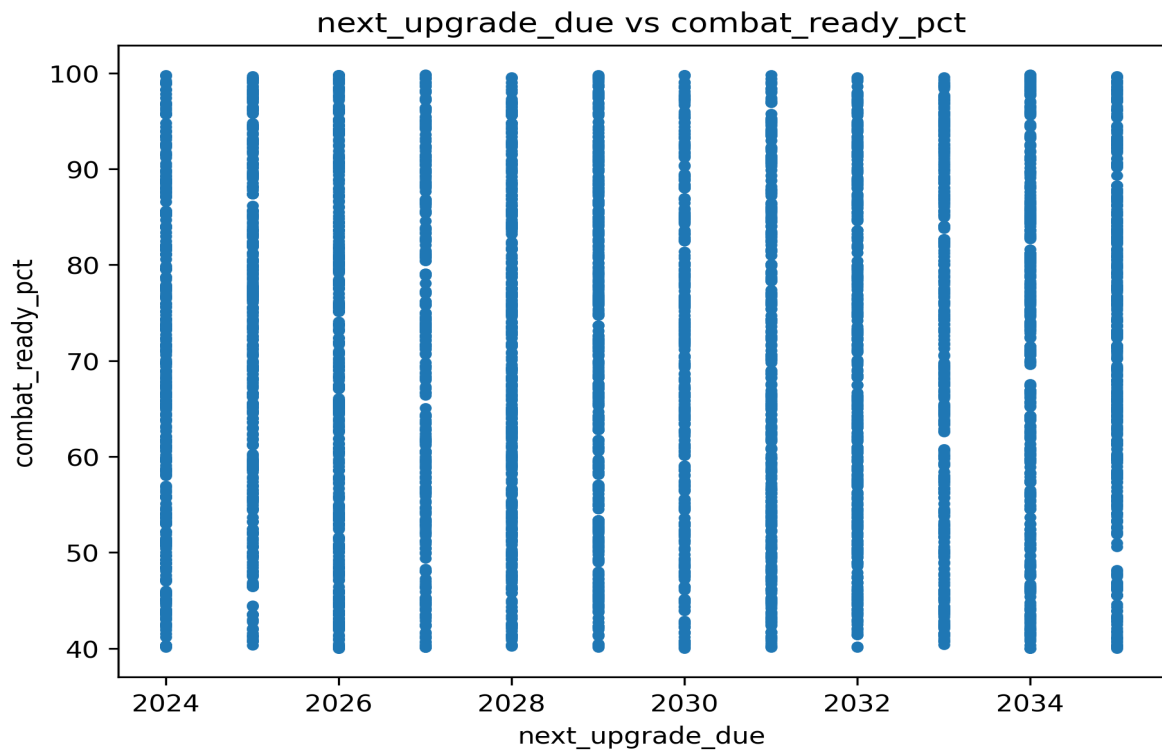
Distribution of total_value_m_usd.



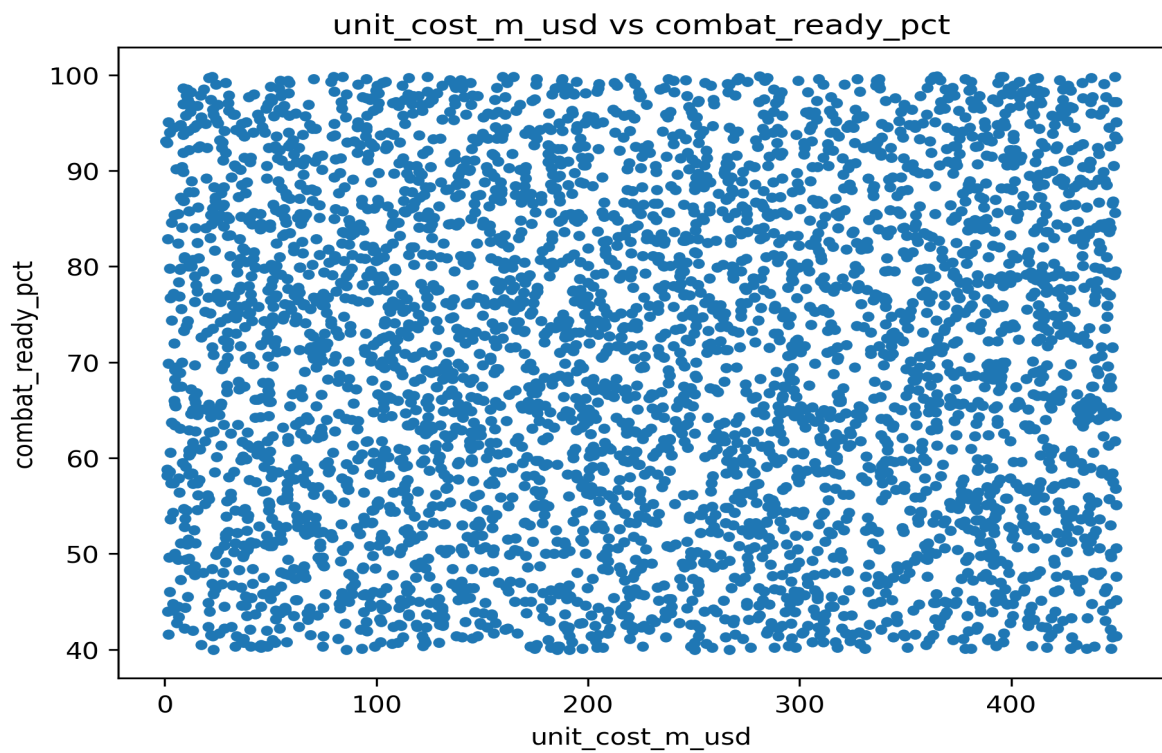
Relationship between units_count and combat_ready_pct.



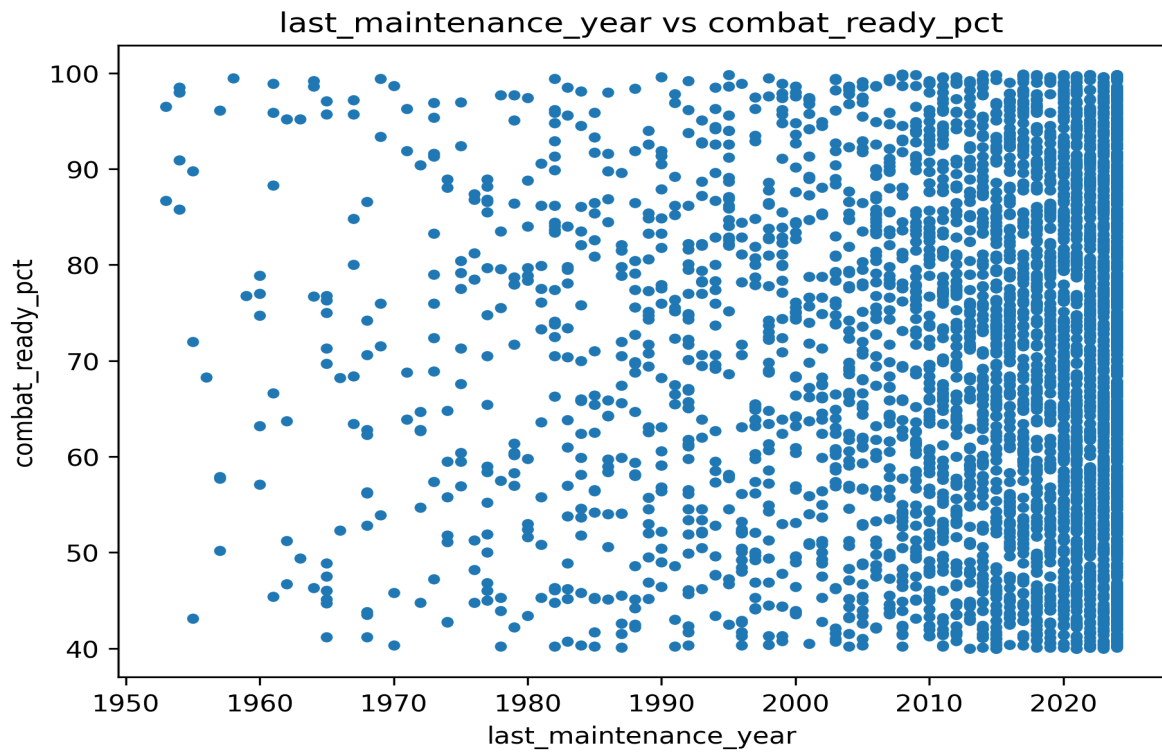
Relationship between total_value_m_usd and combat_ready_pct.



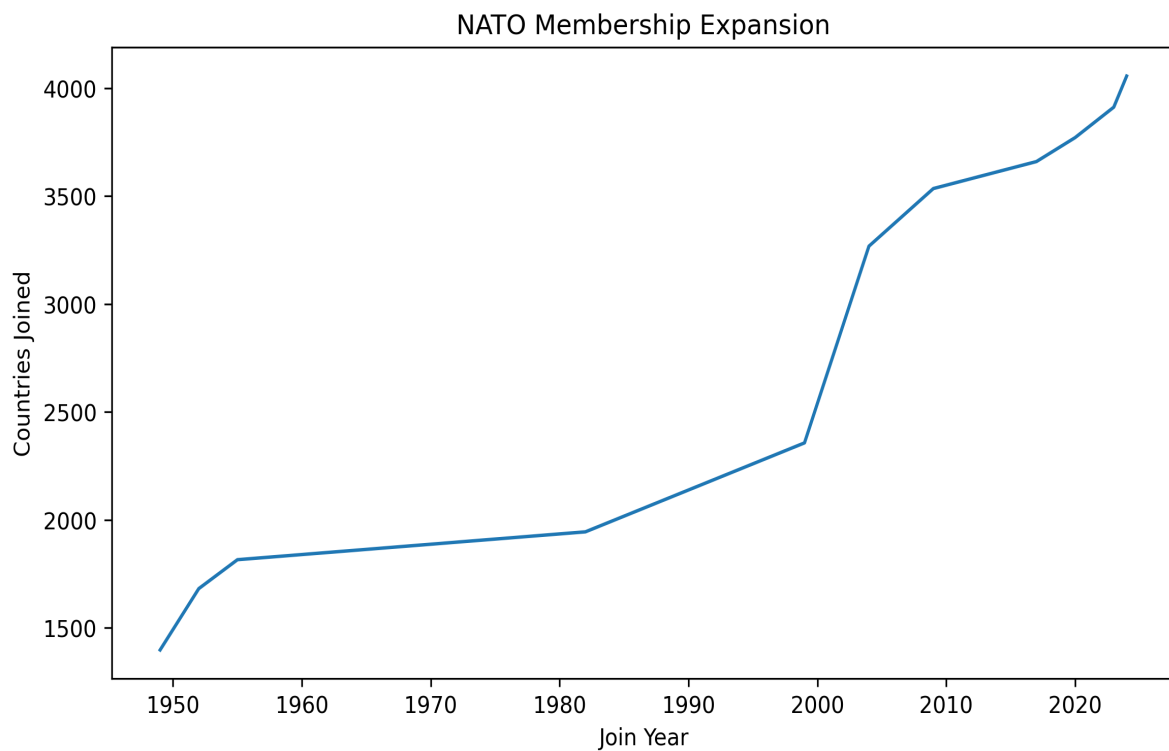
Relationship between next_upgrade_due and combat_ready_pct.



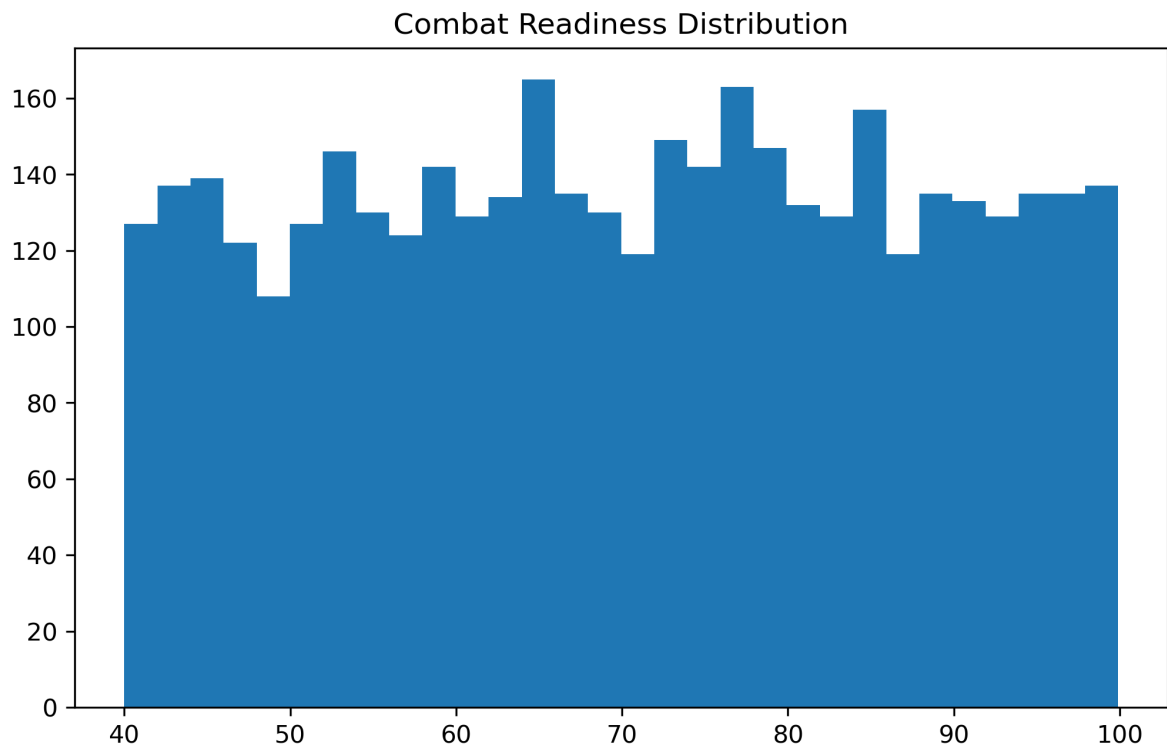
Relationship between unit_cost_m_usd and combat_ready_pct.



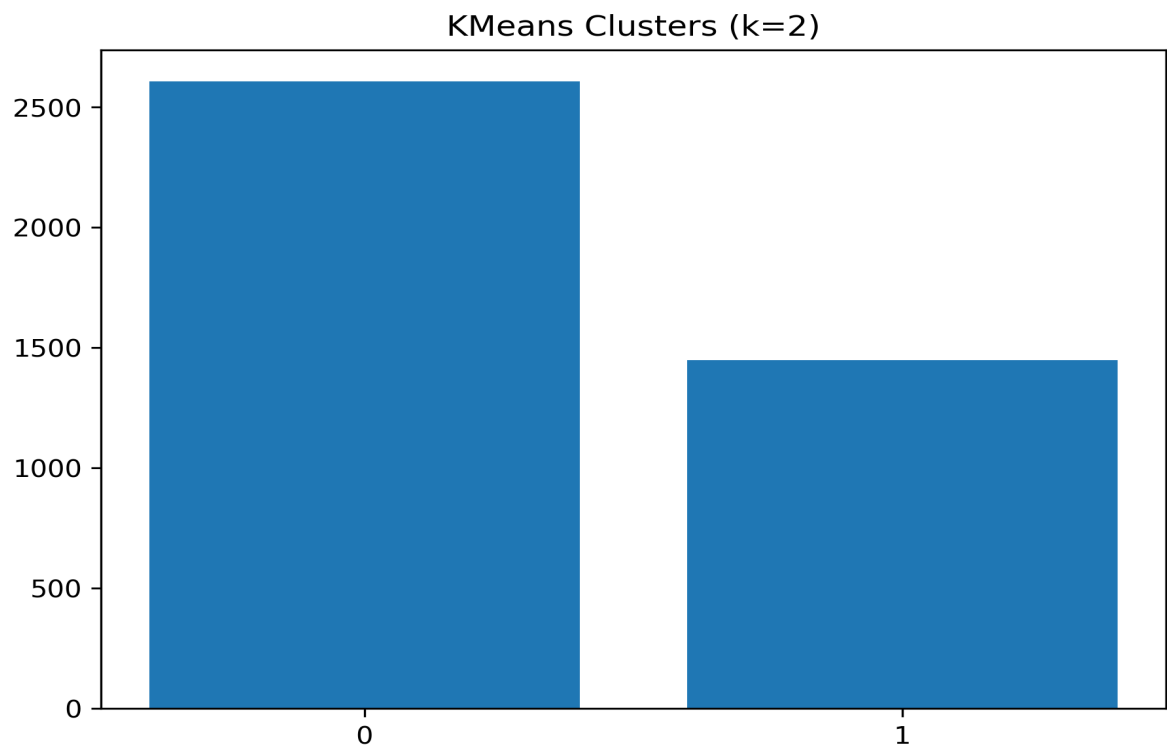
Relationship between last_maintenance_year and combat_ready_pct.



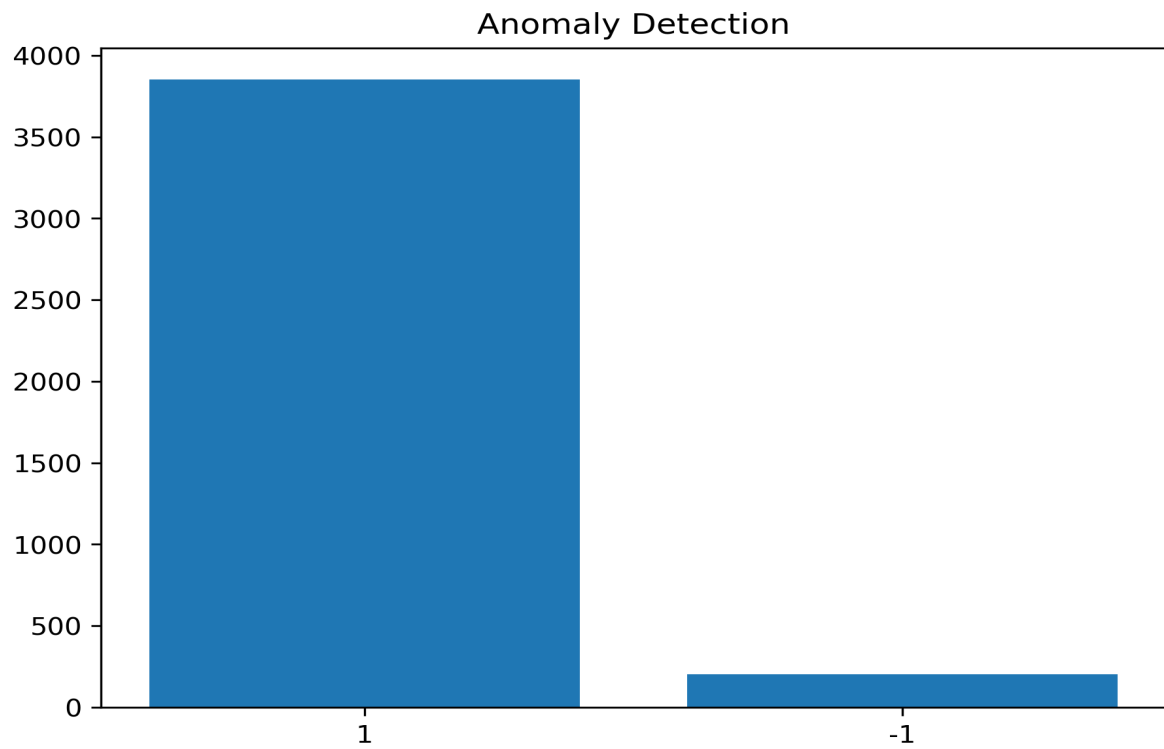
Timeline of NATO membership growth.



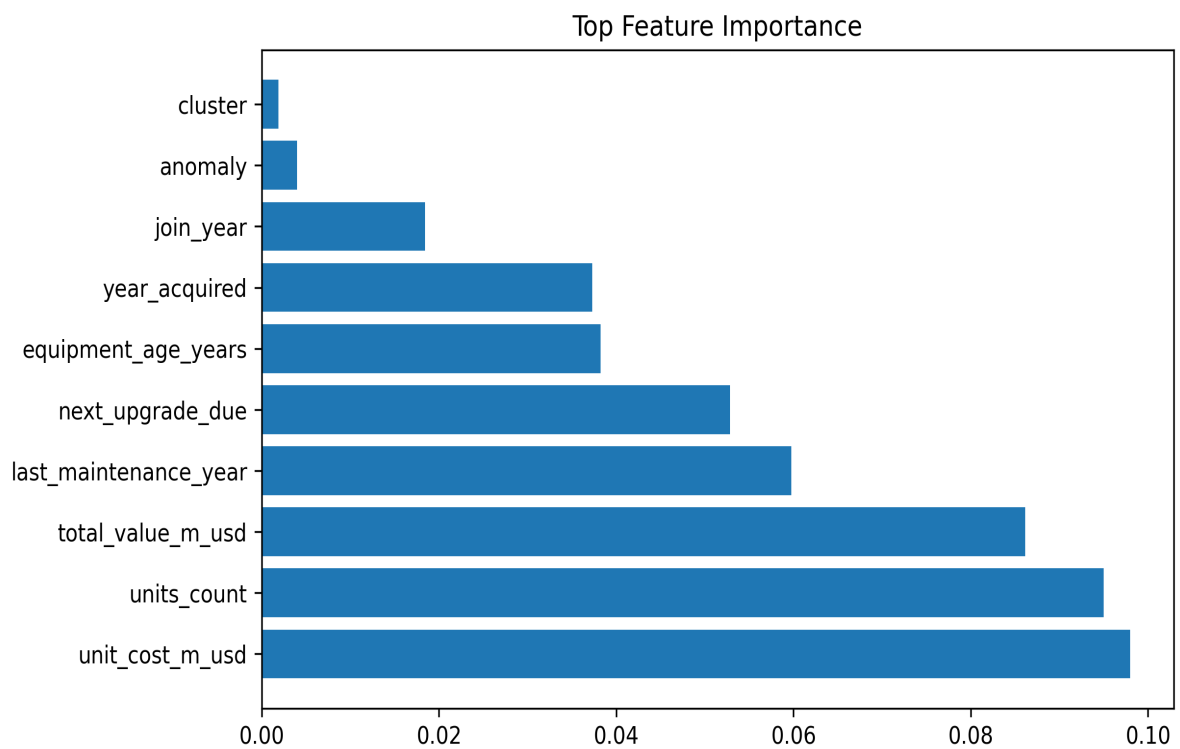
Combat readiness distribution.



Validated KMeans clustering using silhouette optimization (k=2).



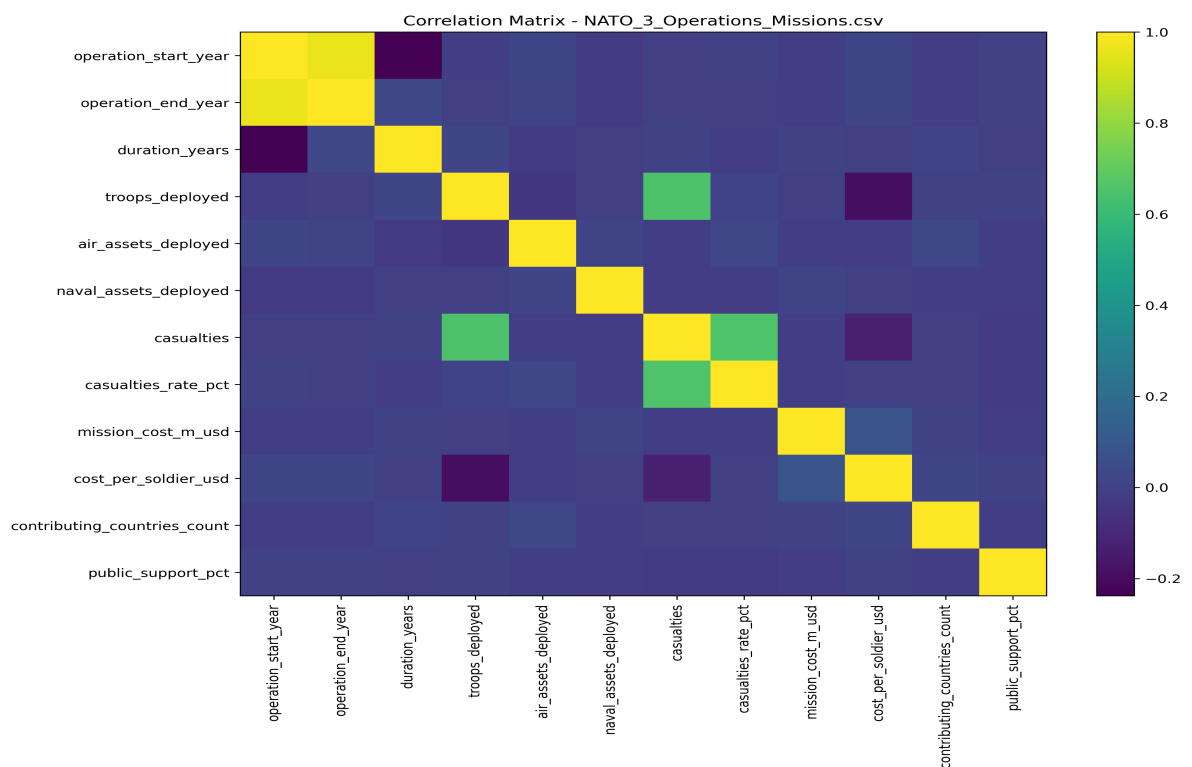
Isolation Forest anomaly detection identifying unusual observations.



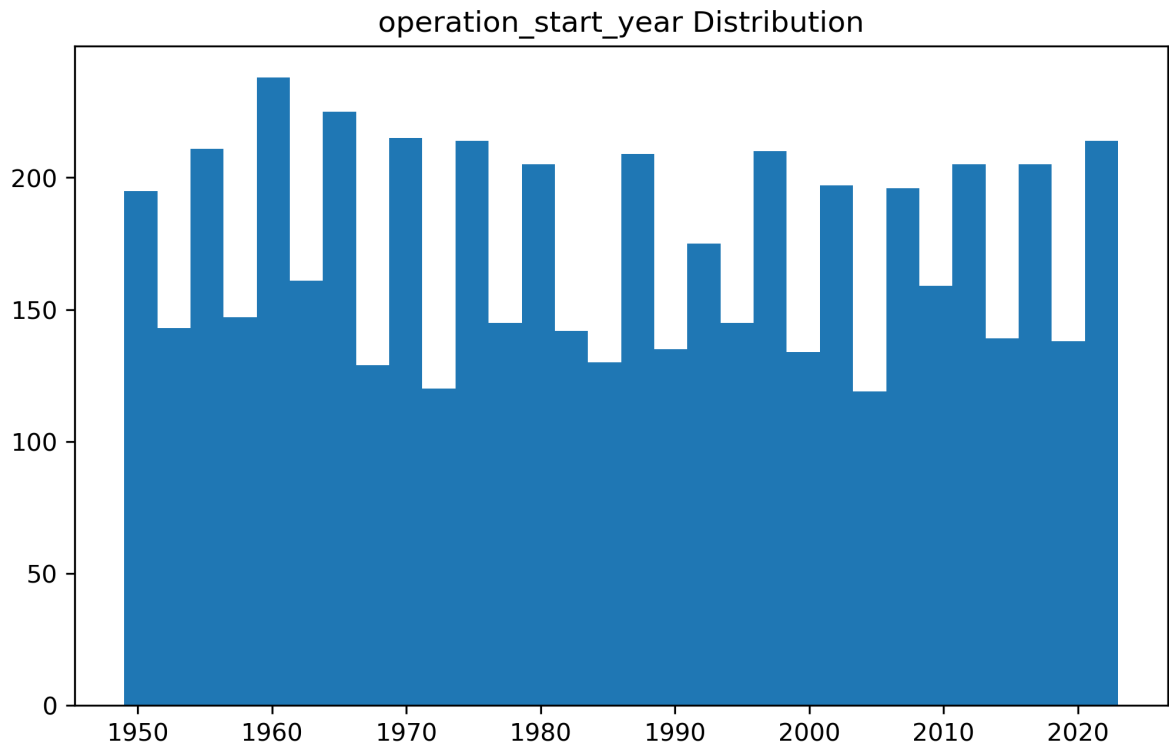
Random Forest feature importance.

Dataset Analysis: NATO_3_Operations_Missions.csv

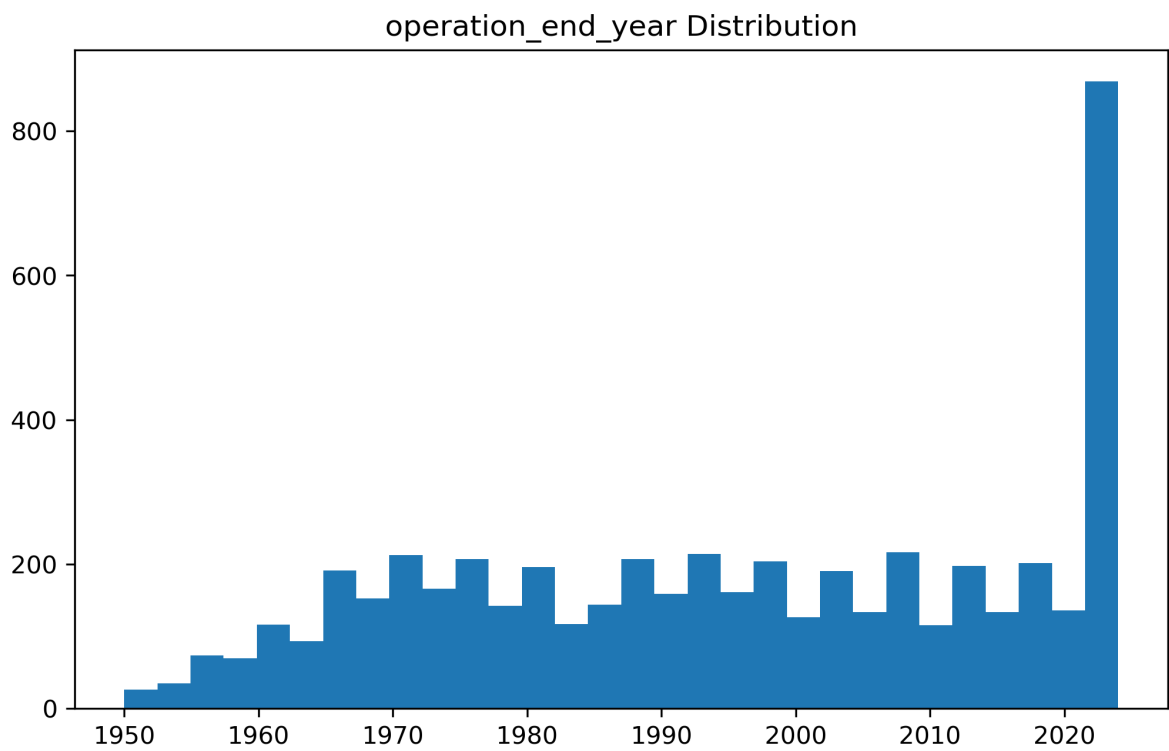
Rows: 5200
Columns: 31
Target Variable: mission_cost_m_usd
Numerical Features: 12
Categorical Features: 17
Generated Charts: 16
Optimal Clusters: 2
Silhouette Score: 0.058
Random Forest RMSE: 392.179
Gradient Boosting RMSE: 544.054



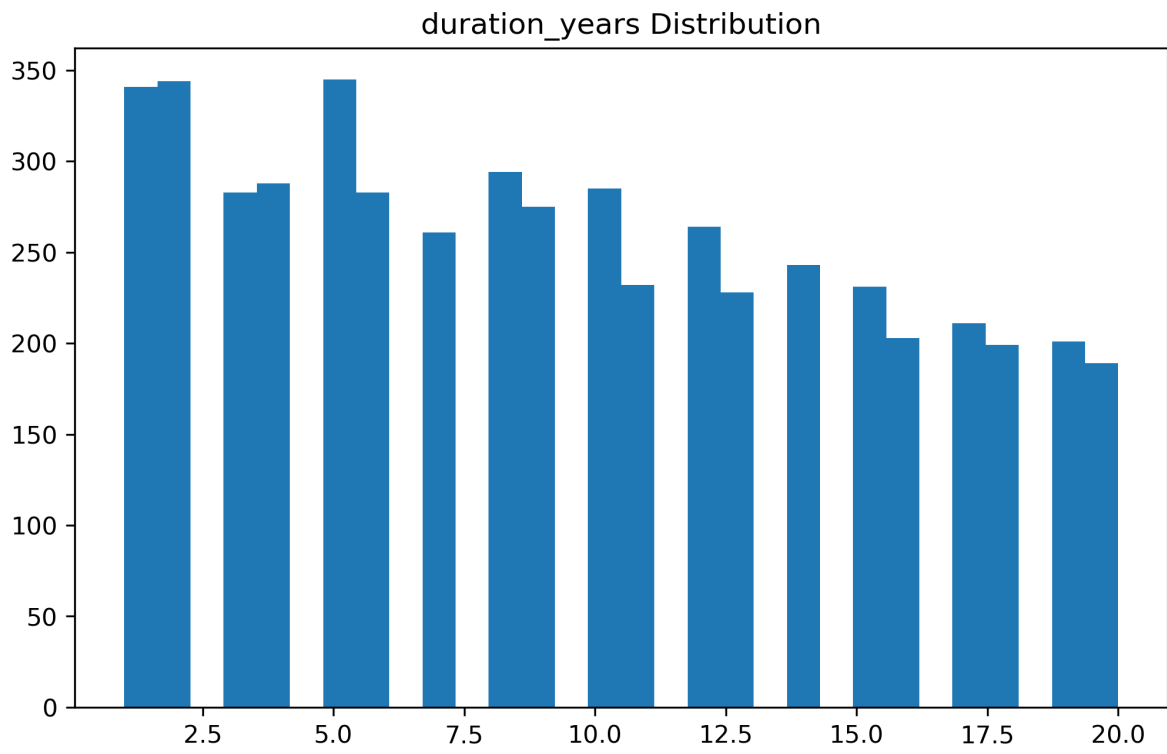
Correlation matrix of valid numerical variables only.



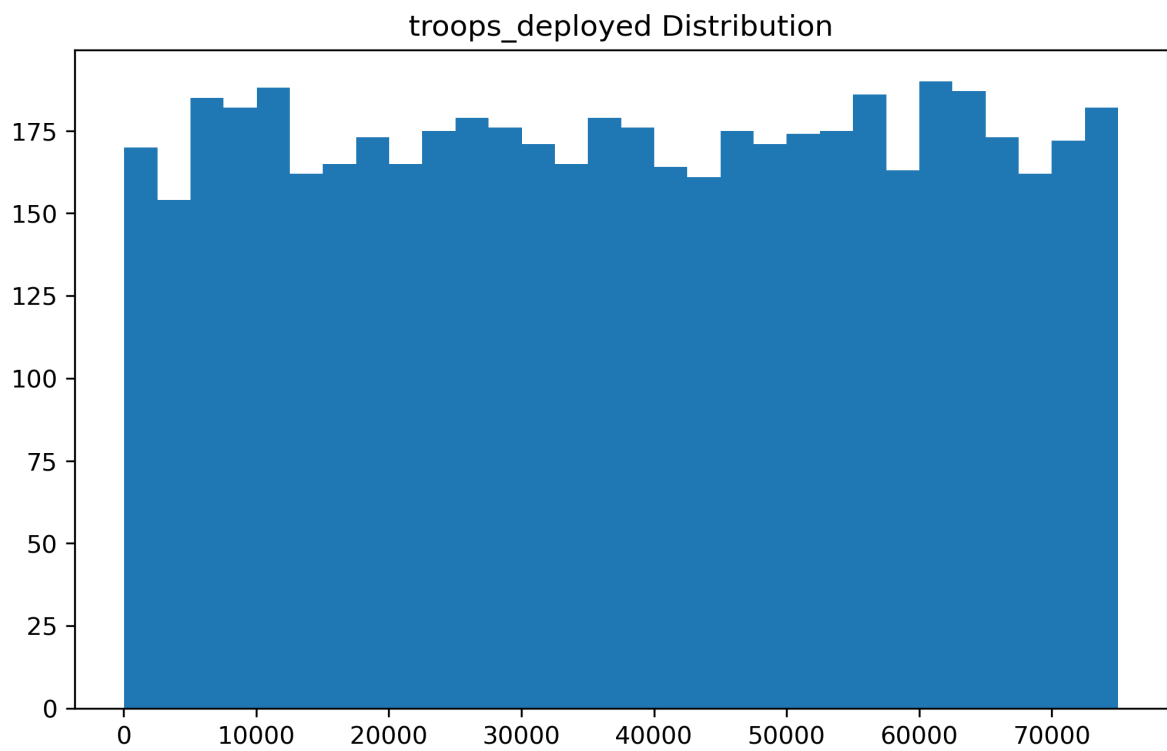
Distribution of operation_start_year.



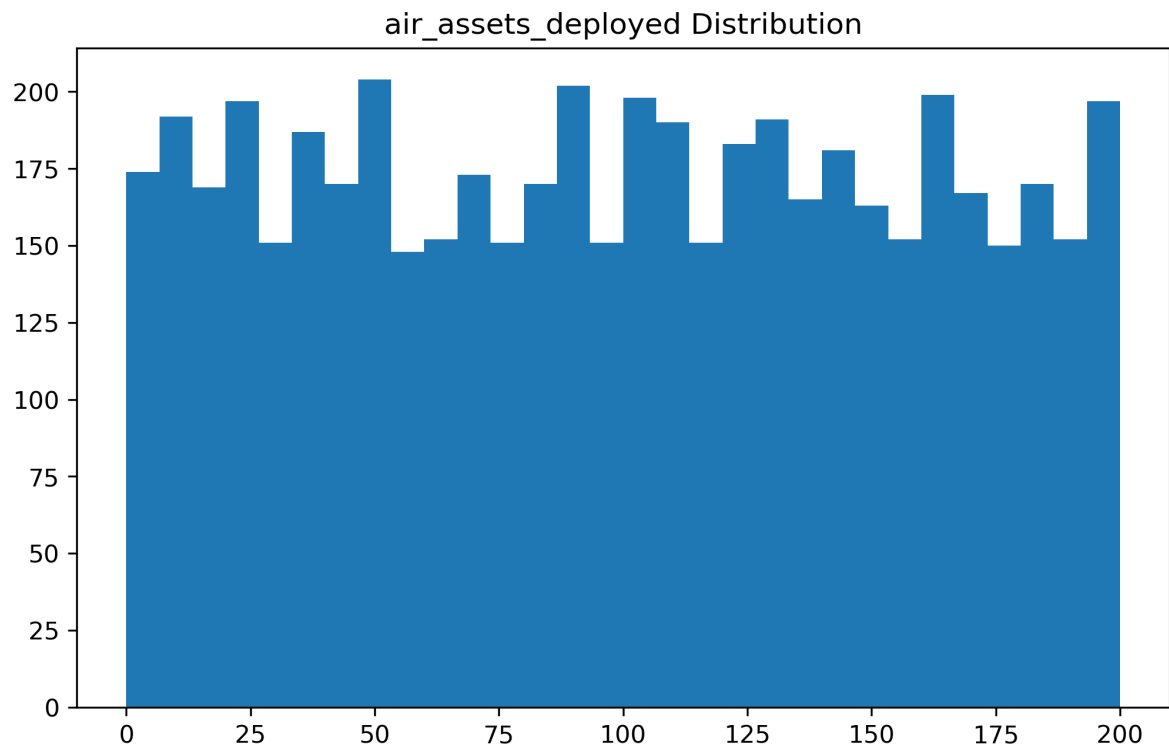
Distribution of operation_end_year.



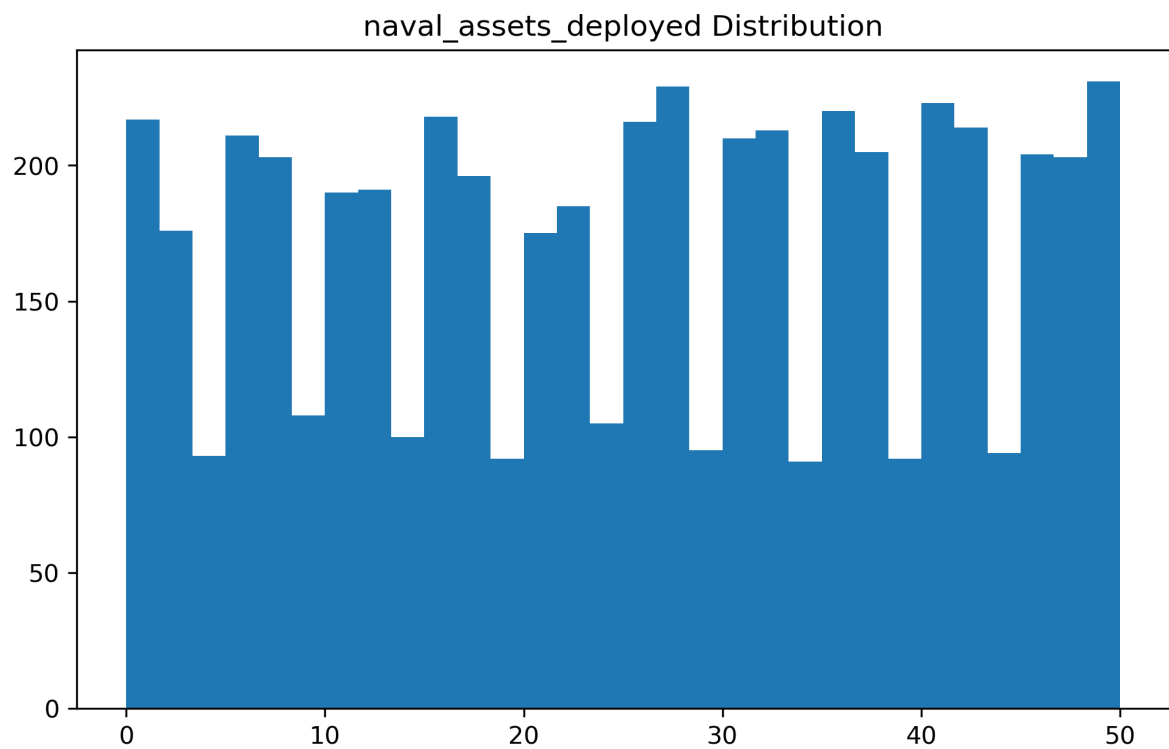
Distribution of duration_years.



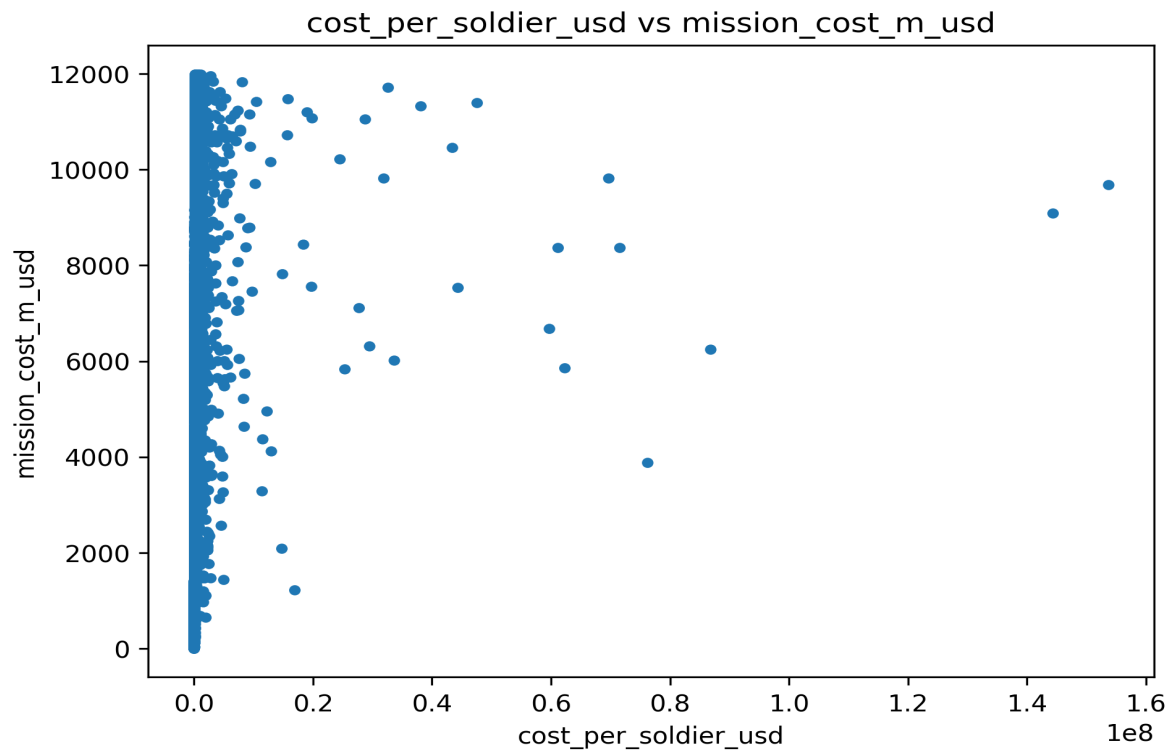
Distribution of troops_deployed.



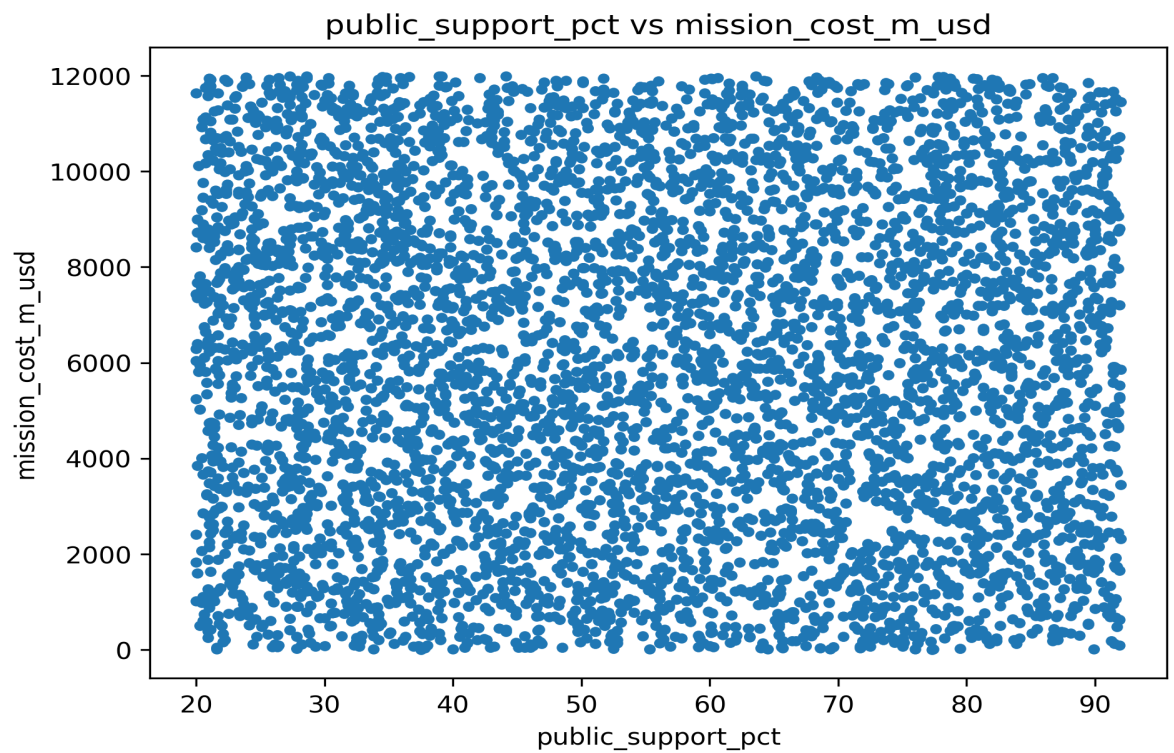
Distribution of air_assets_deployed.



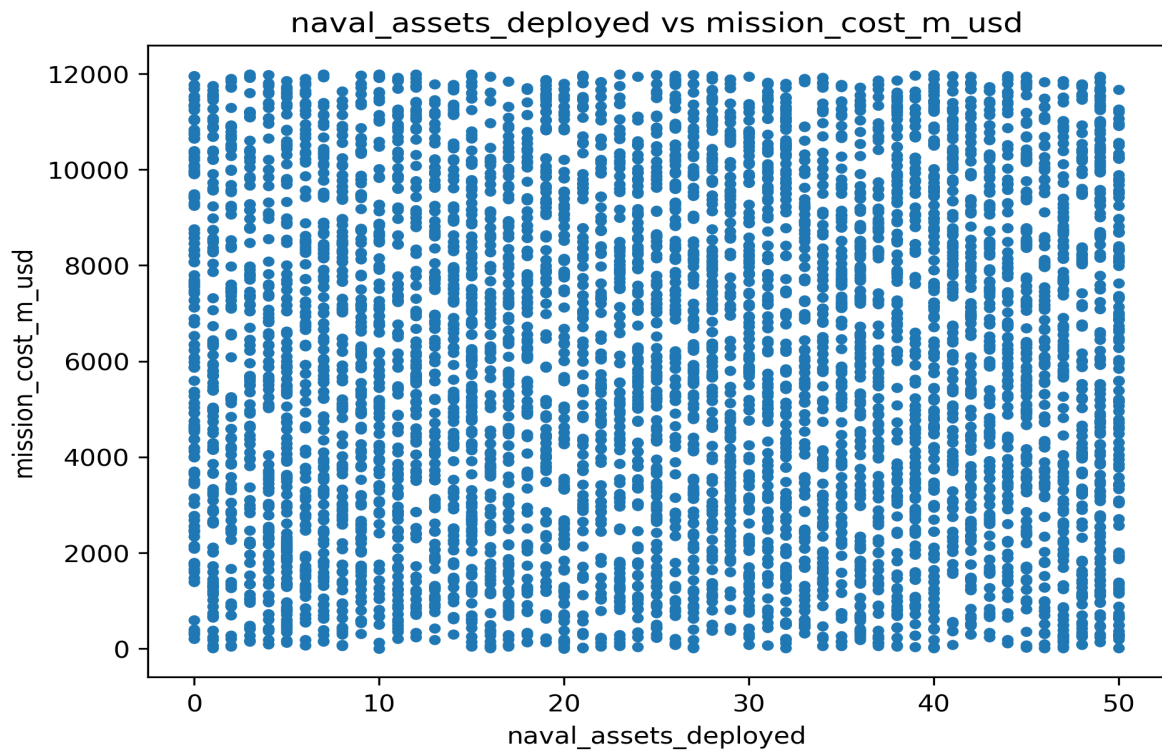
Distribution of naval_assets_deployed.



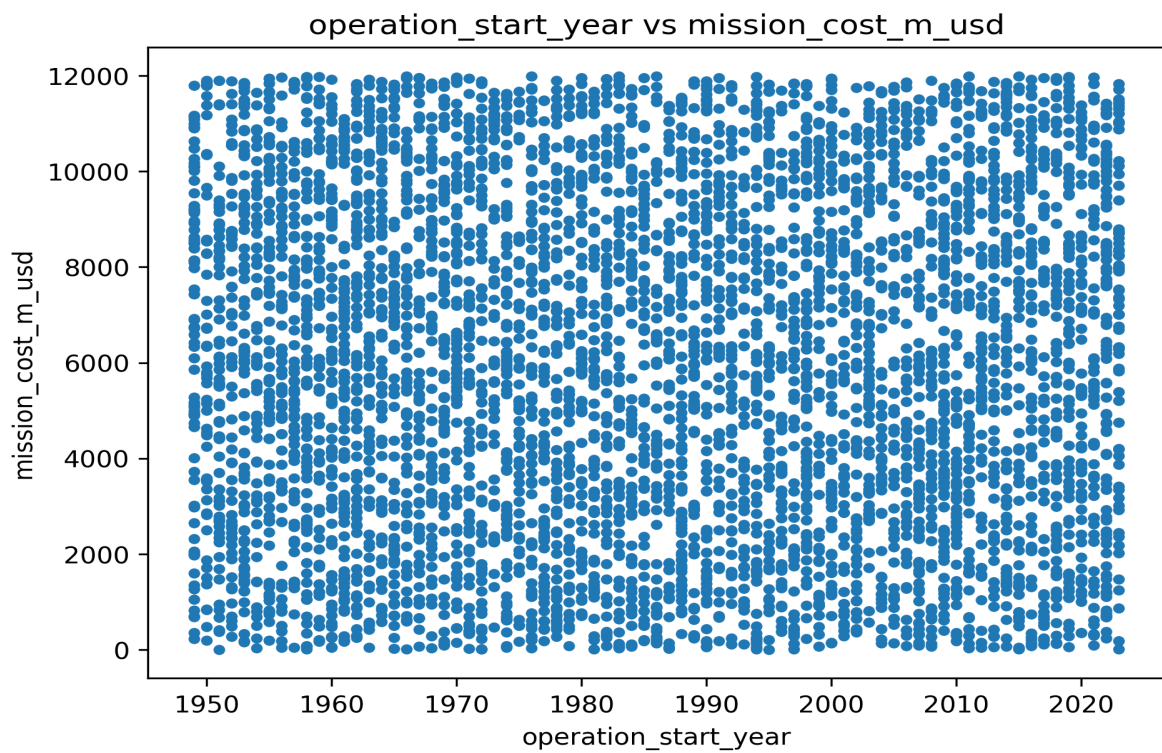
Relationship between cost_per_soldier_usd and mission_cost_m_usd.



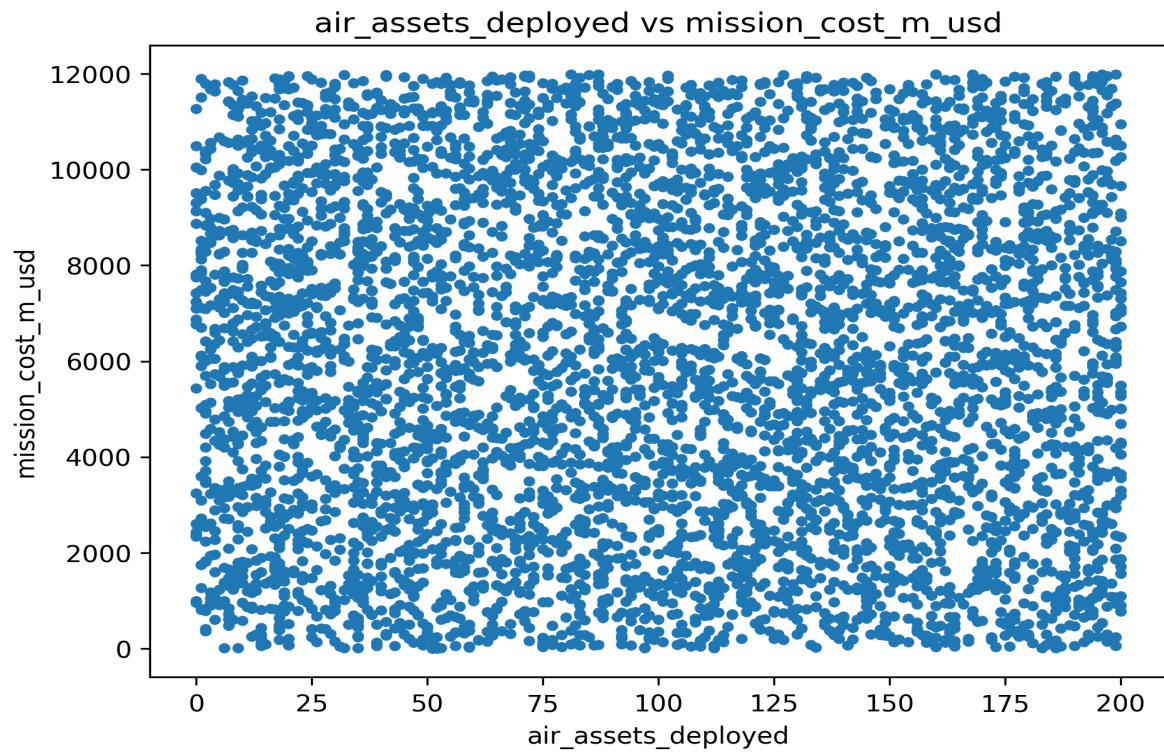
Relationship between public_support_pct and mission_cost_m_usd.



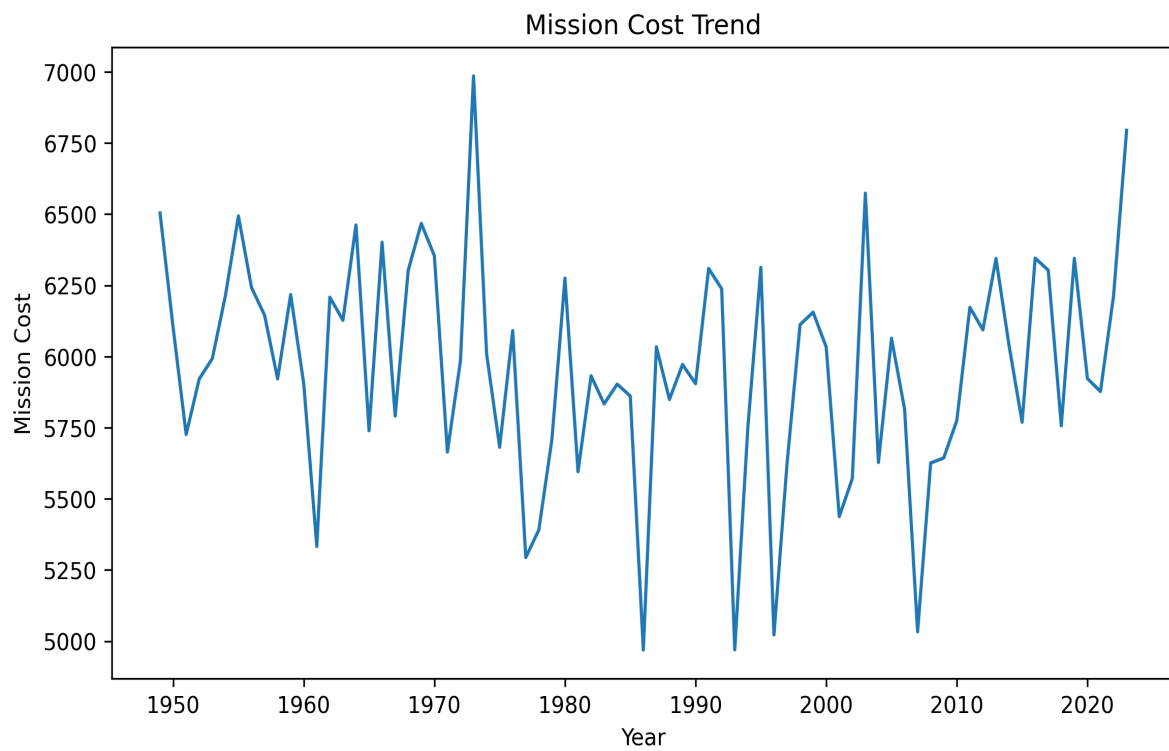
Relationship between naval_assets_deployed and mission_cost_m_usd.



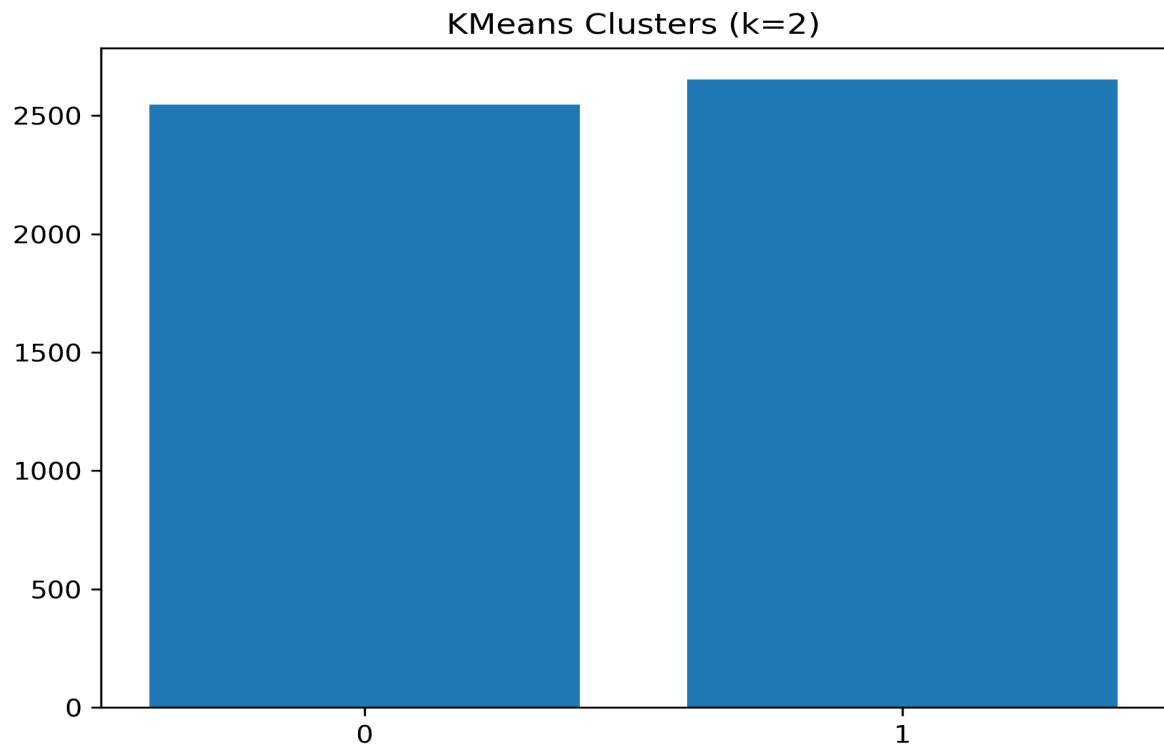
Relationship between operation_start_year and mission_cost_m_usd.



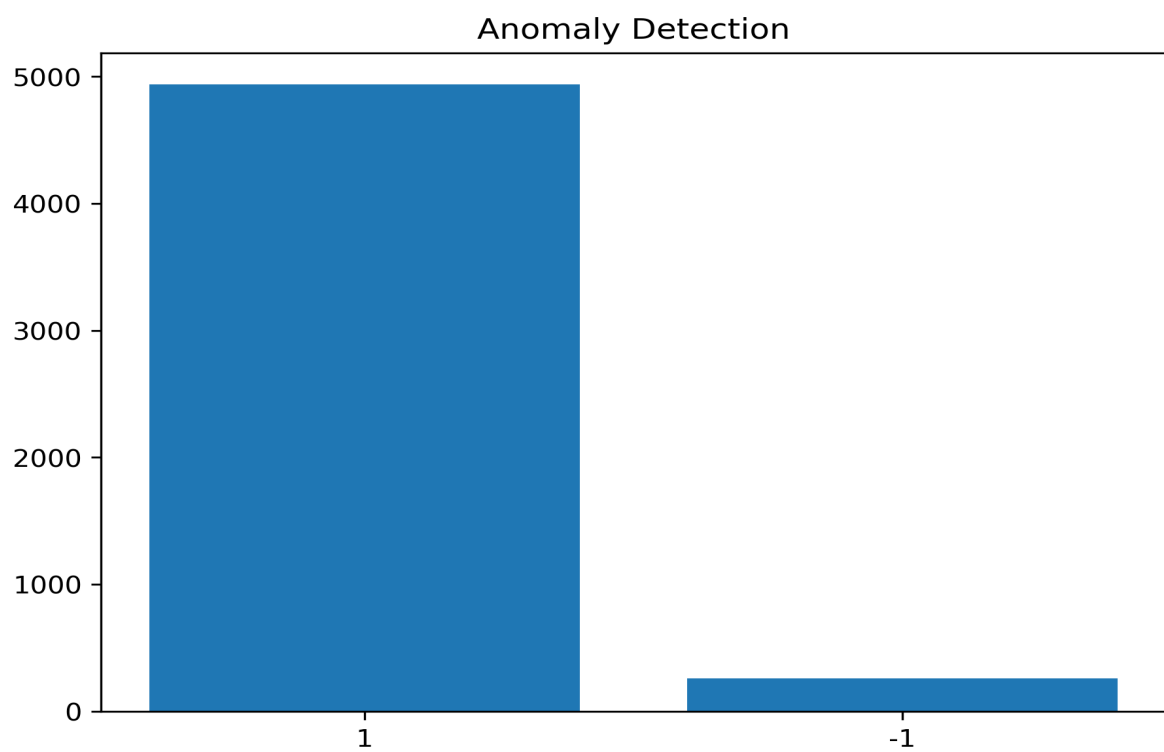
Relationship between air_assets_deployed and mission_cost_m_usd.



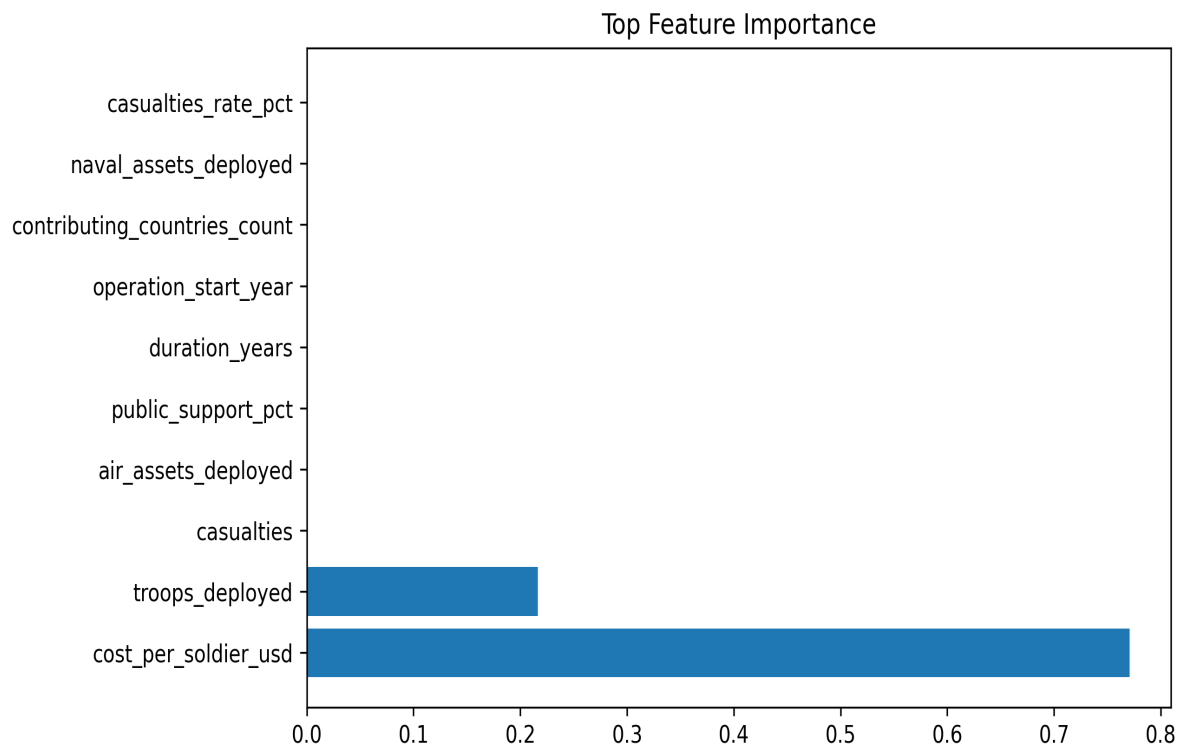
Average NATO mission cost over time.



Validated KMeans clustering using silhouette optimization (k=2).



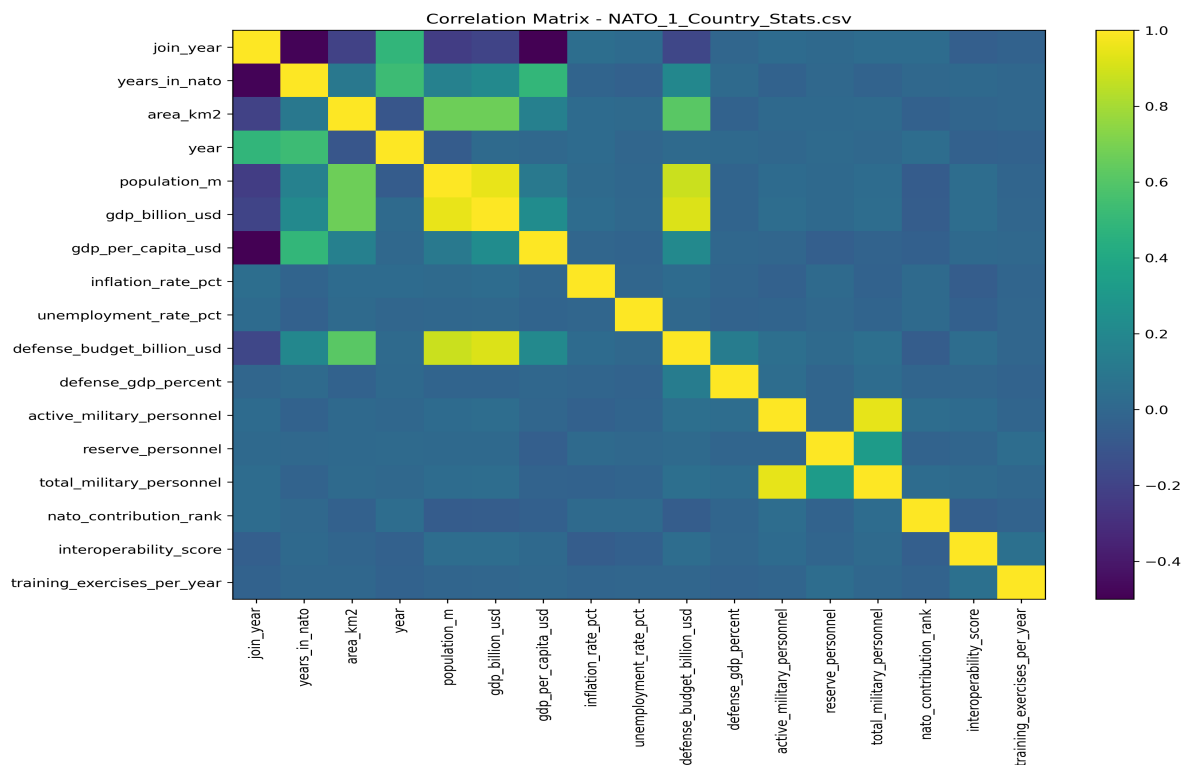
Isolation Forest anomaly detection identifying unusual observations.



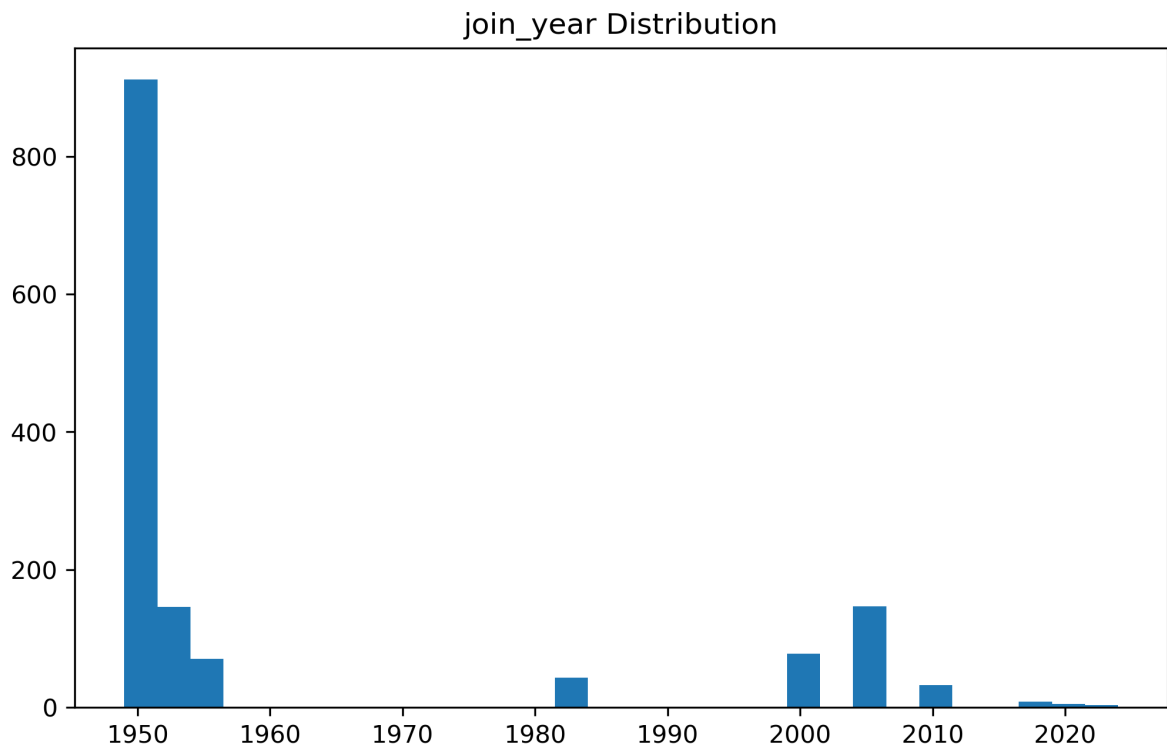
Random Forest feature importance.

Dataset Analysis: NATO_1_Country_Stats.csv

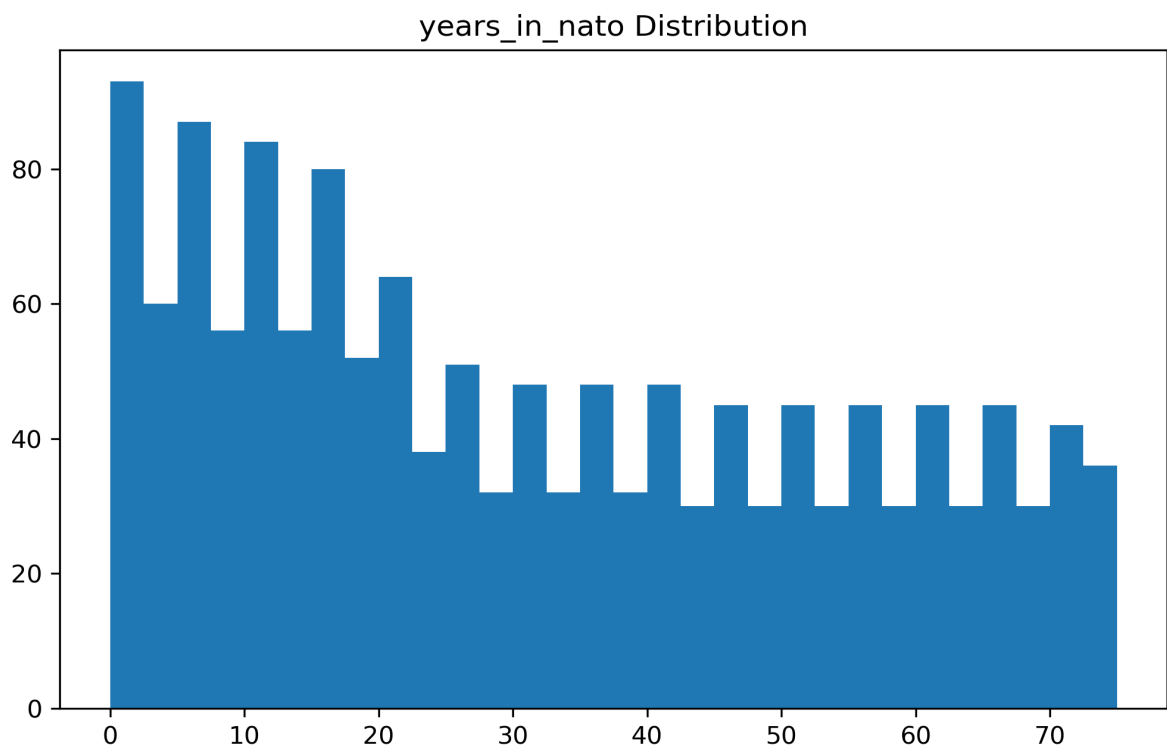
Rows: 1444
Columns: 28
Target Variable: defense_budget_billion_usd
Numerical Features: 17
Categorical Features: 9
Generated Charts: 18
Optimal Clusters: 2
Silhouette Score: 0.373
Random Forest RMSE: 57.312
Gradient Boosting RMSE: 47.03



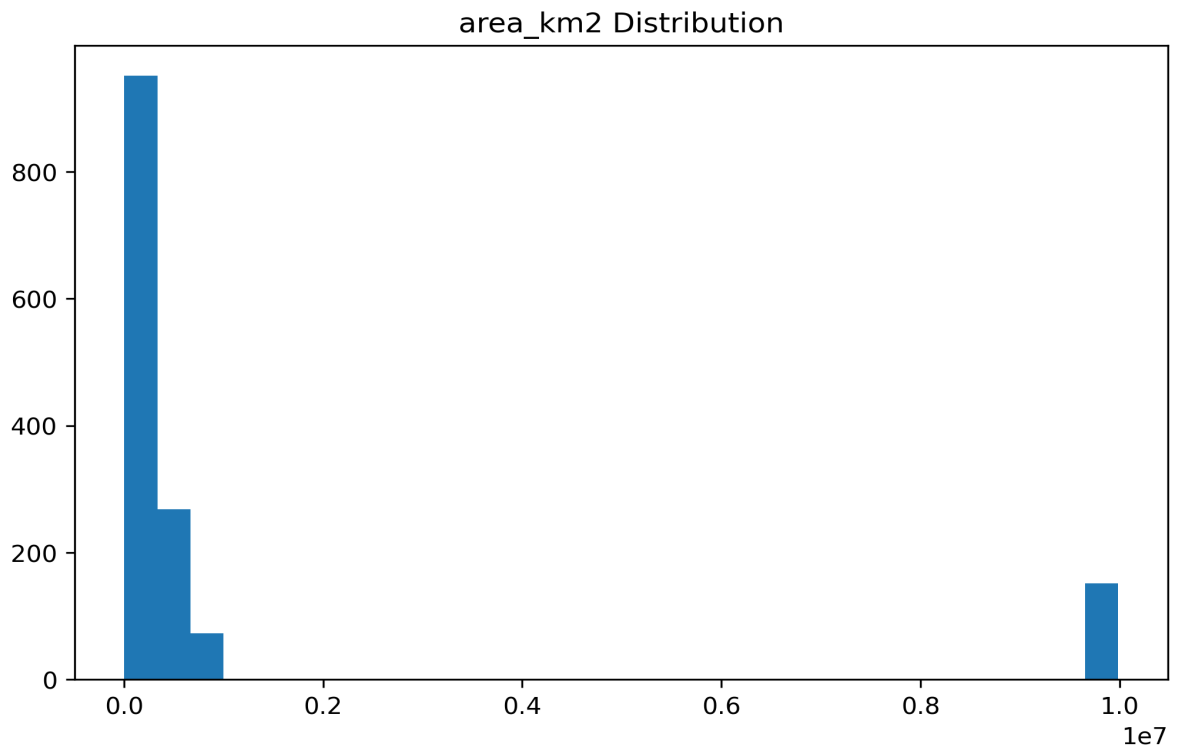
Correlation matrix of valid numerical variables only.



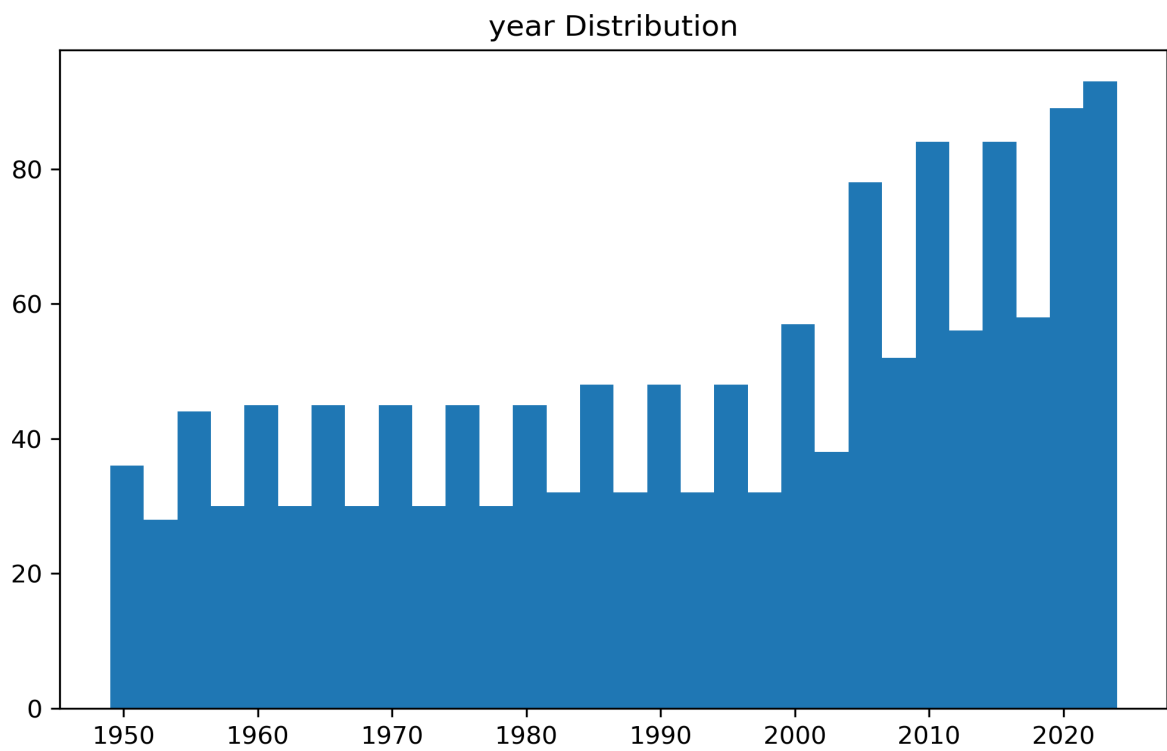
Distribution of join_year.



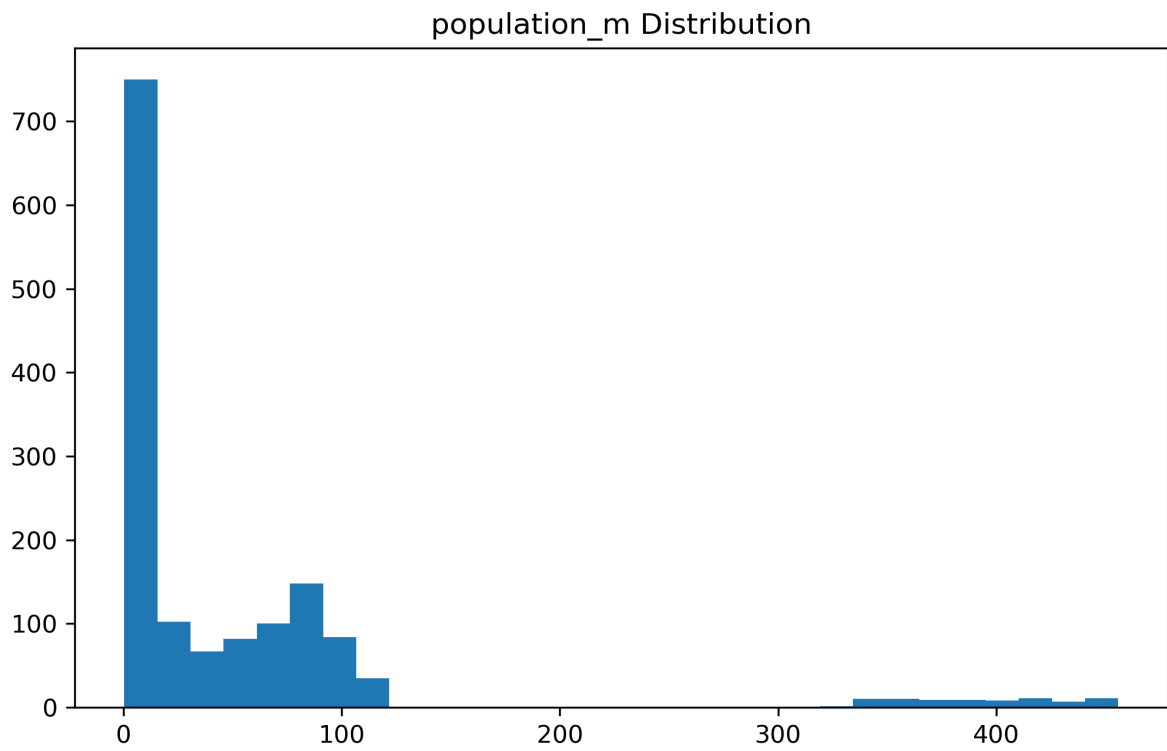
Distribution of years_in_nato.



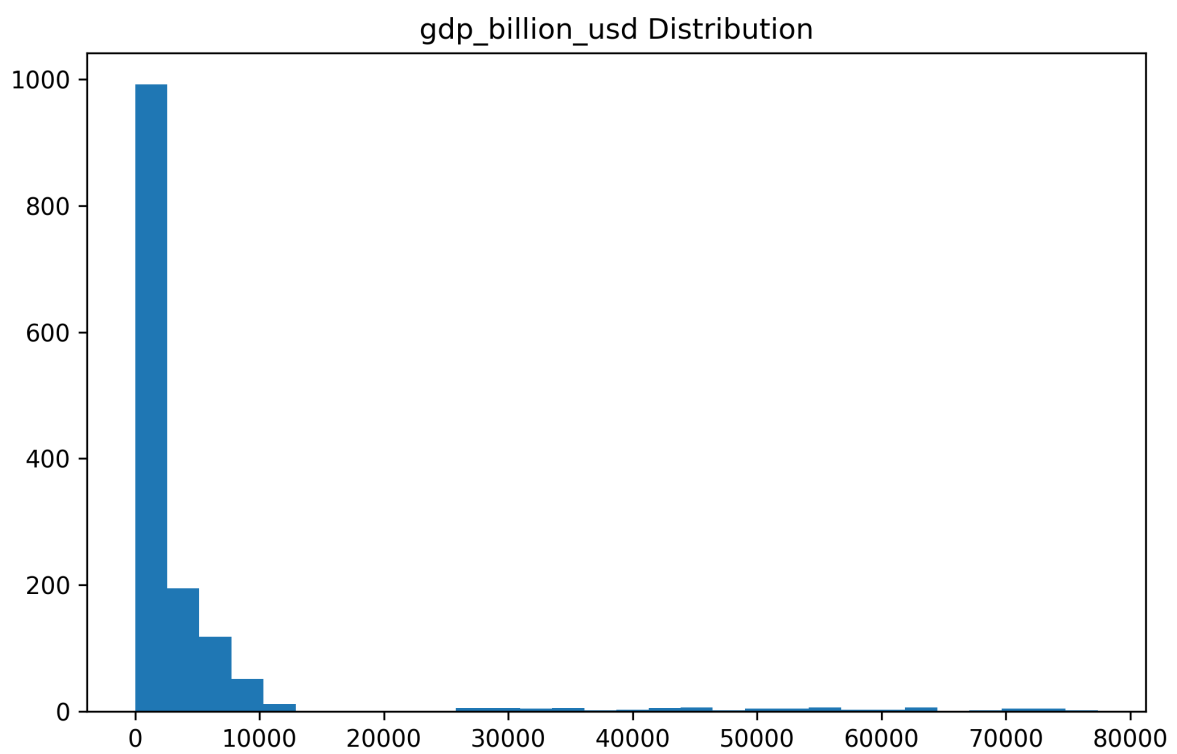
Distribution of area_km2.



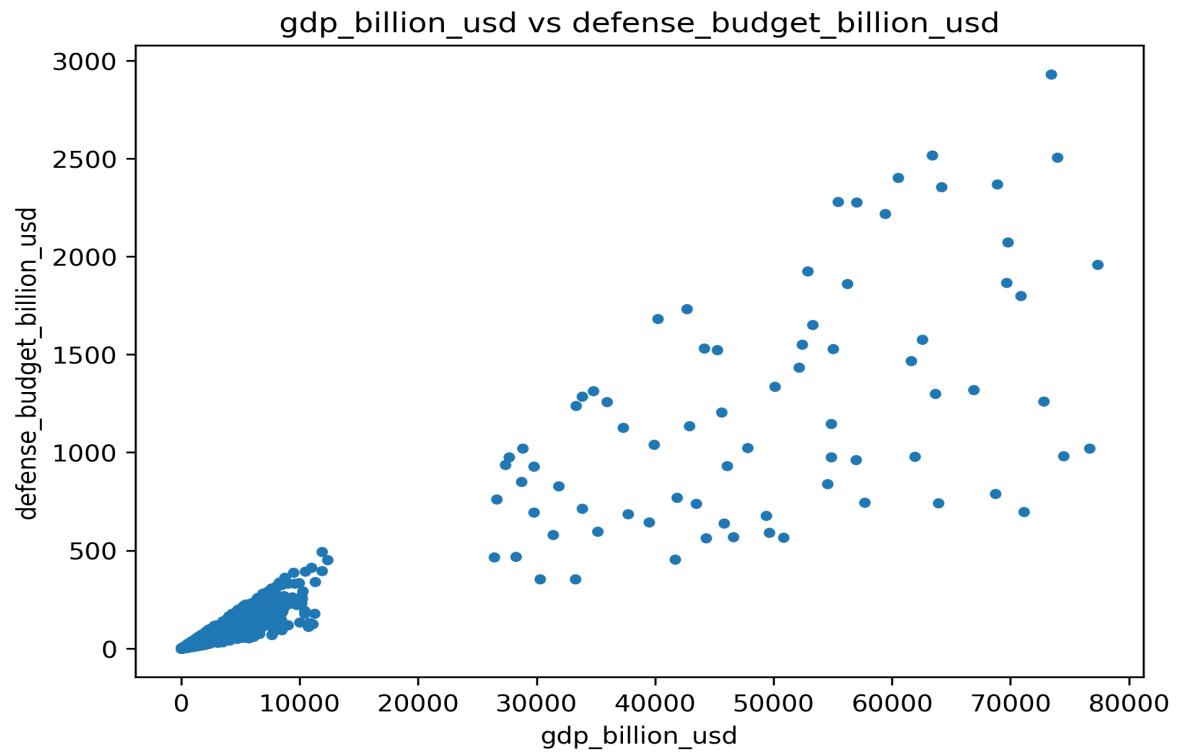
Distribution of year.



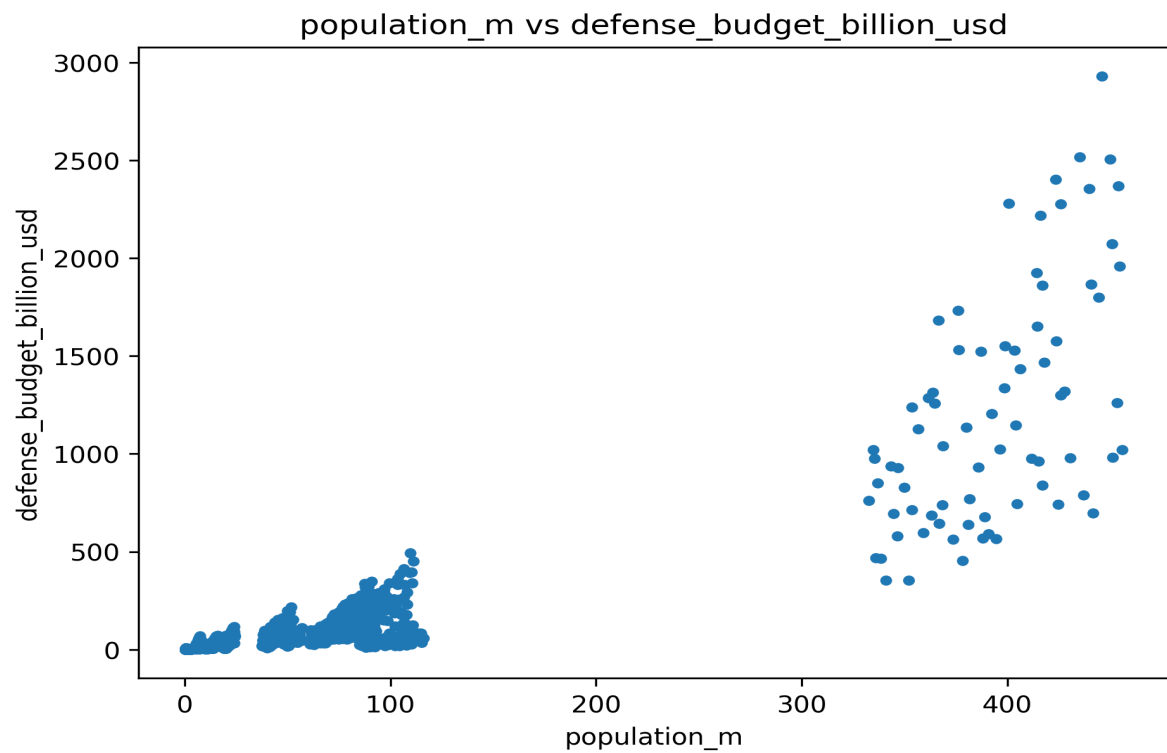
Distribution of population_m.



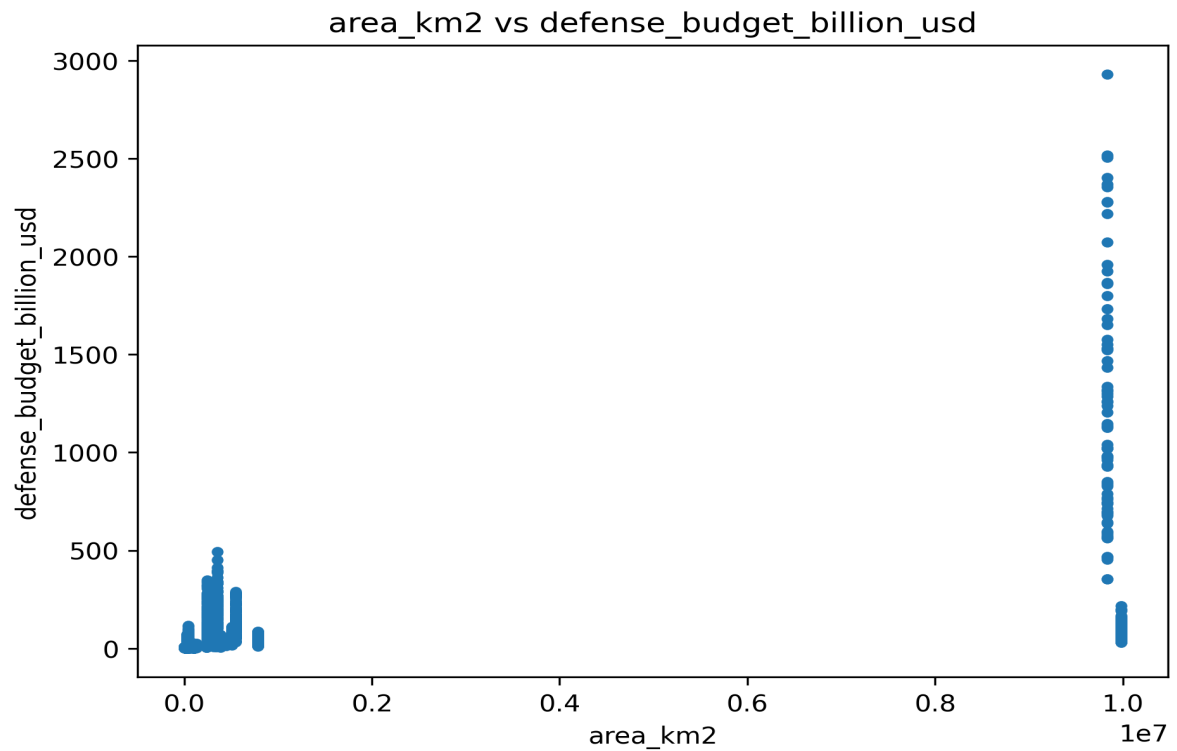
Distribution of gdp_billion_usd.



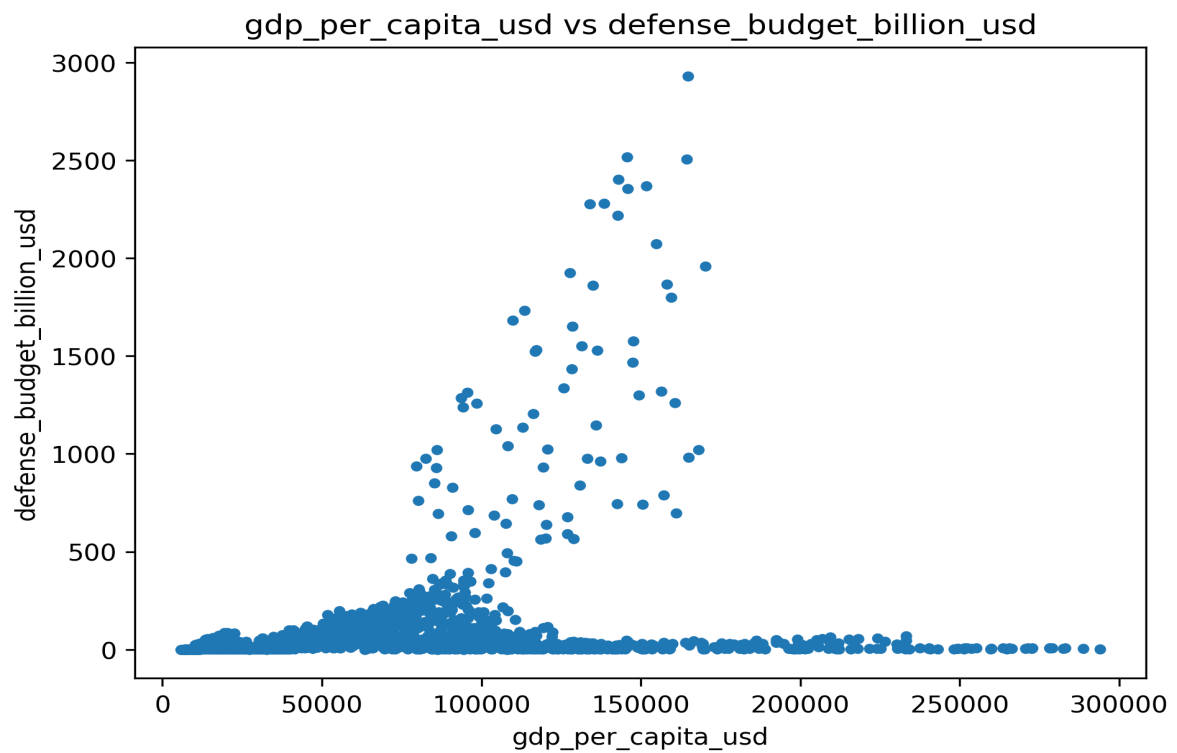
Relationship between gdp_billion_usd and defense_budget_billion_usd.



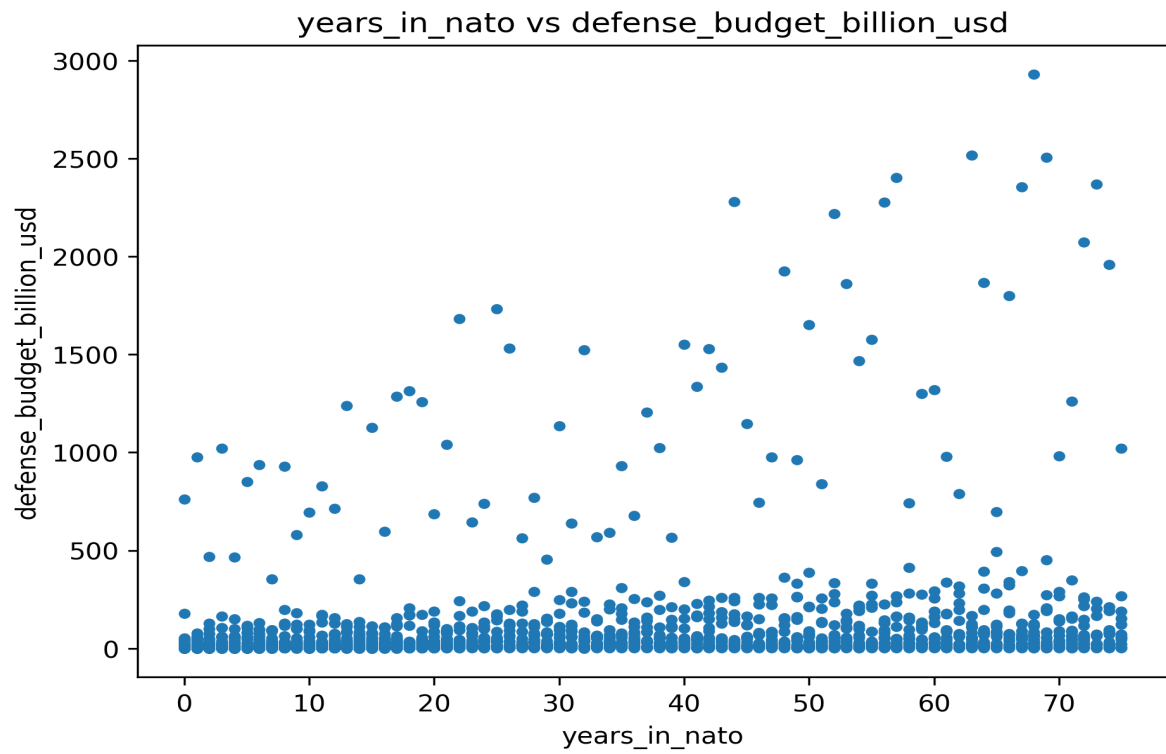
Relationship between population_m and defense_budget_billion_usd.



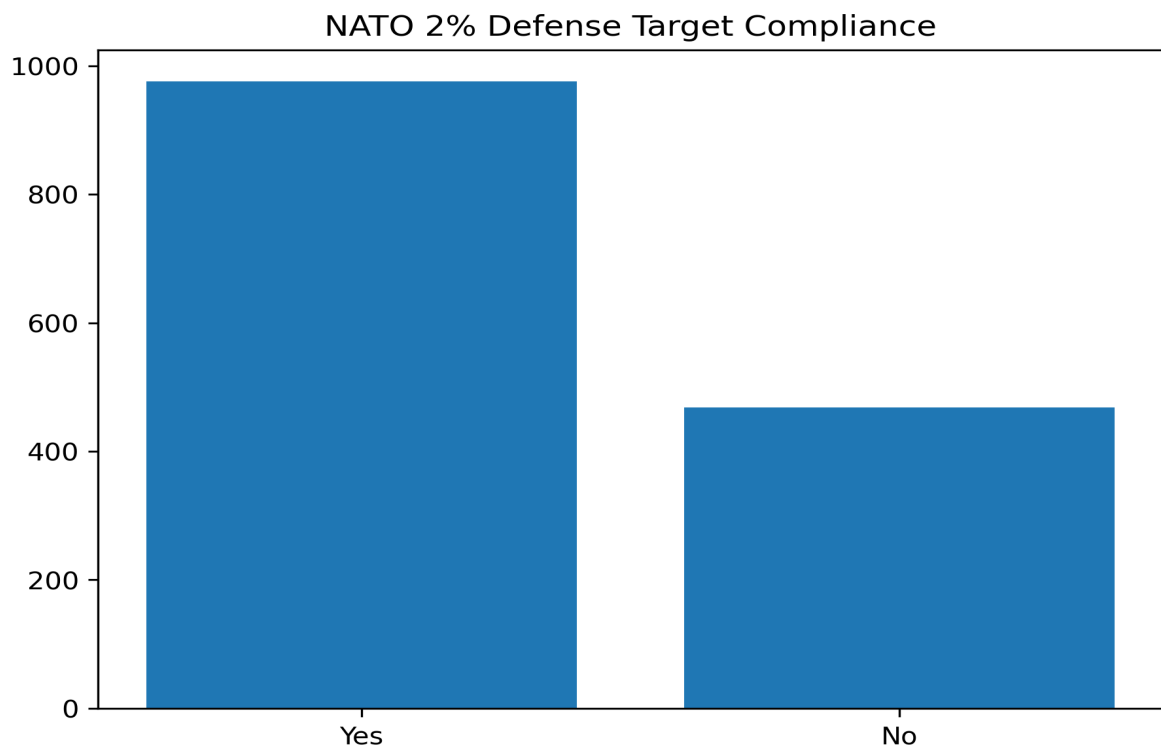
Relationship between area_km2 and defense_budget_billion_usd.



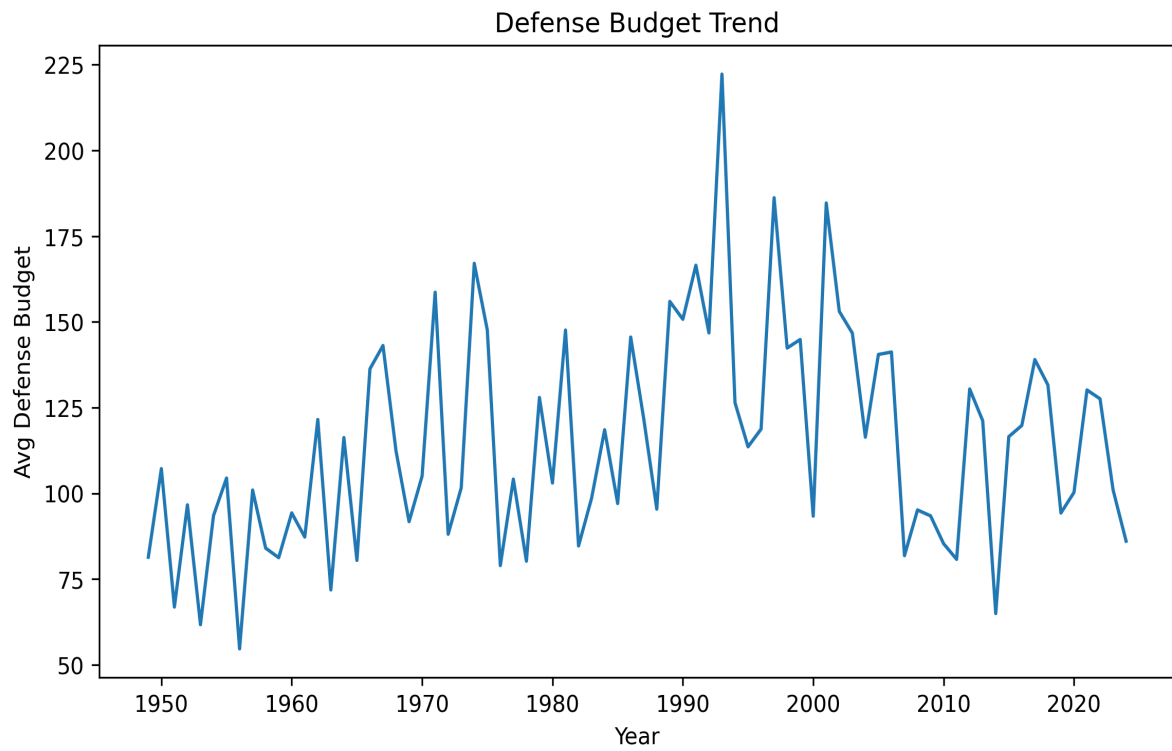
Relationship between gdp_per_capita_usd and defense_budget_billion_usd.



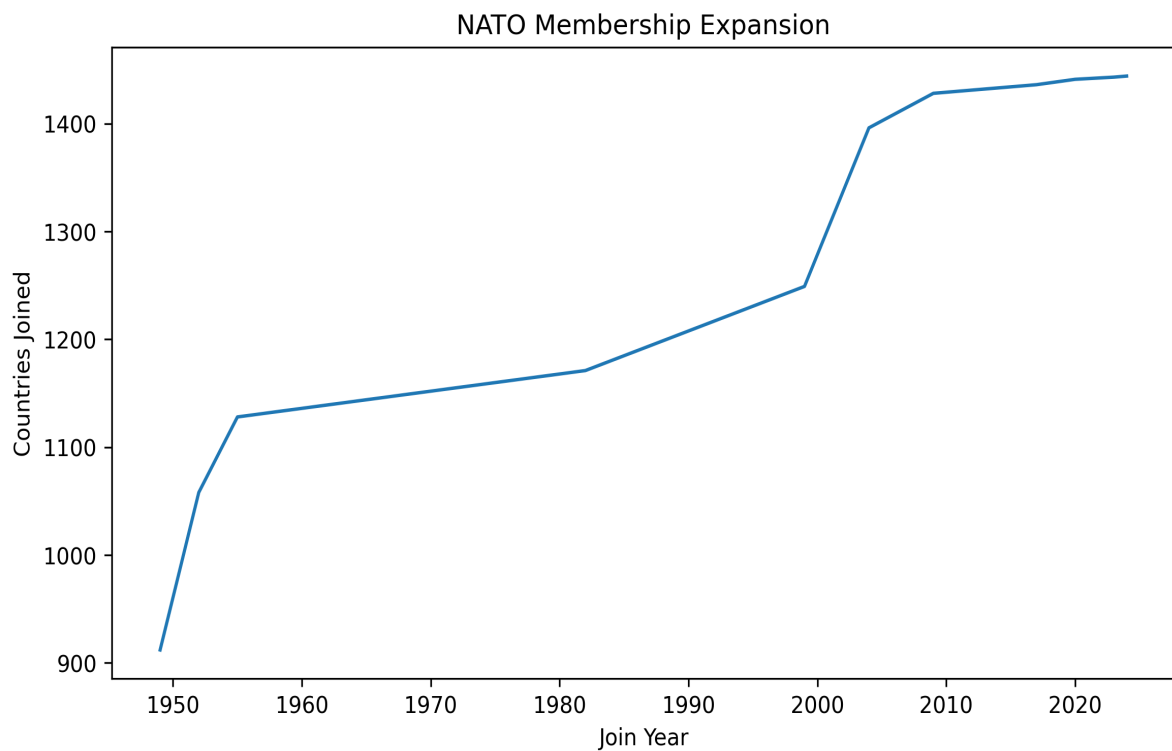
Relationship between years_in_nato and defense_budget_billion_usd.



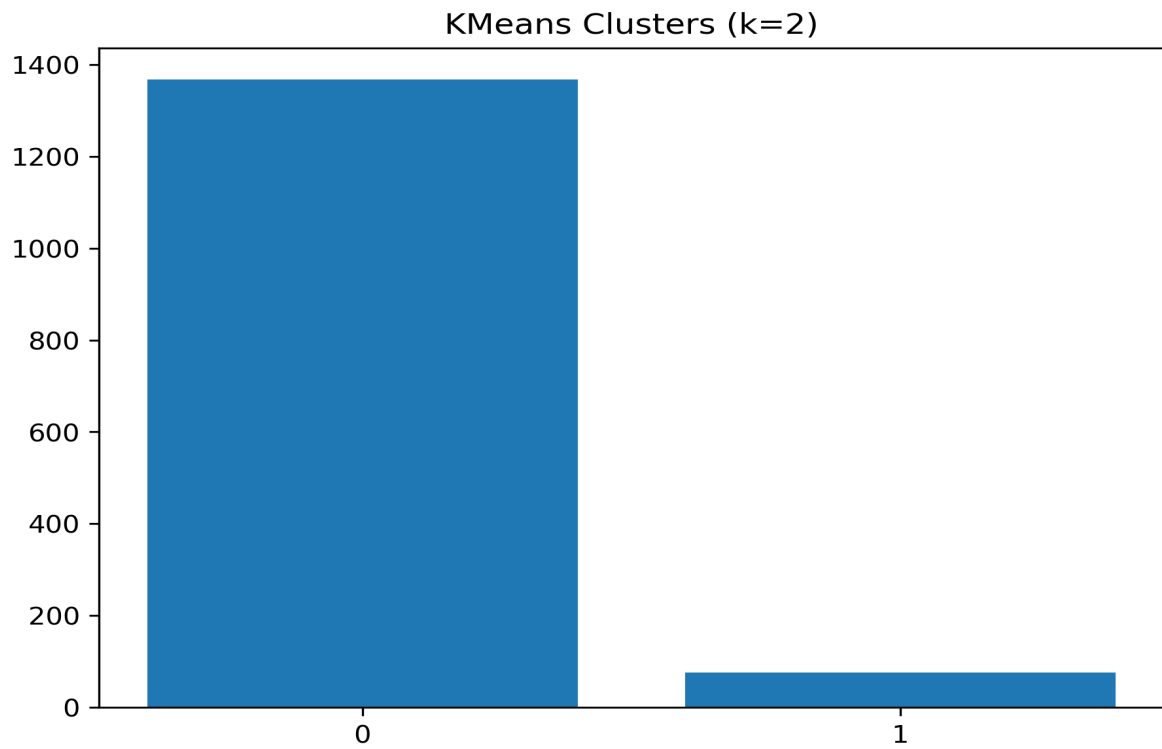
Countries meeting NATO's 2% GDP defense target.



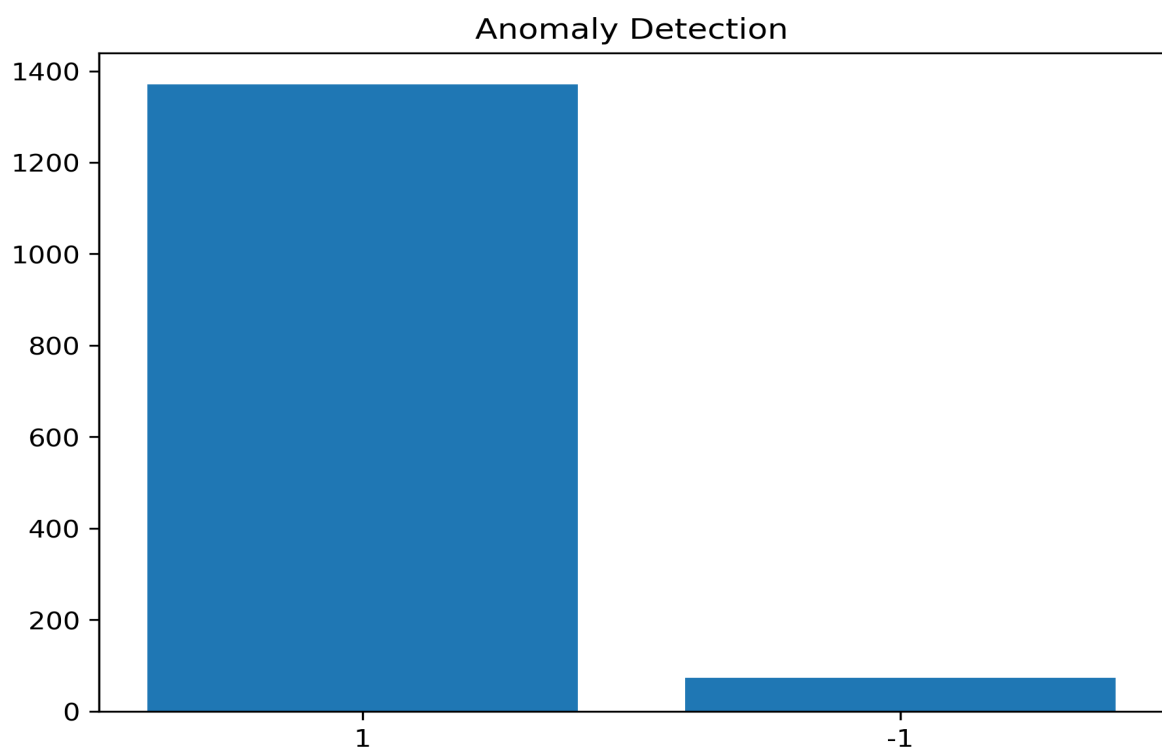
Average NATO defense budget trend over time.



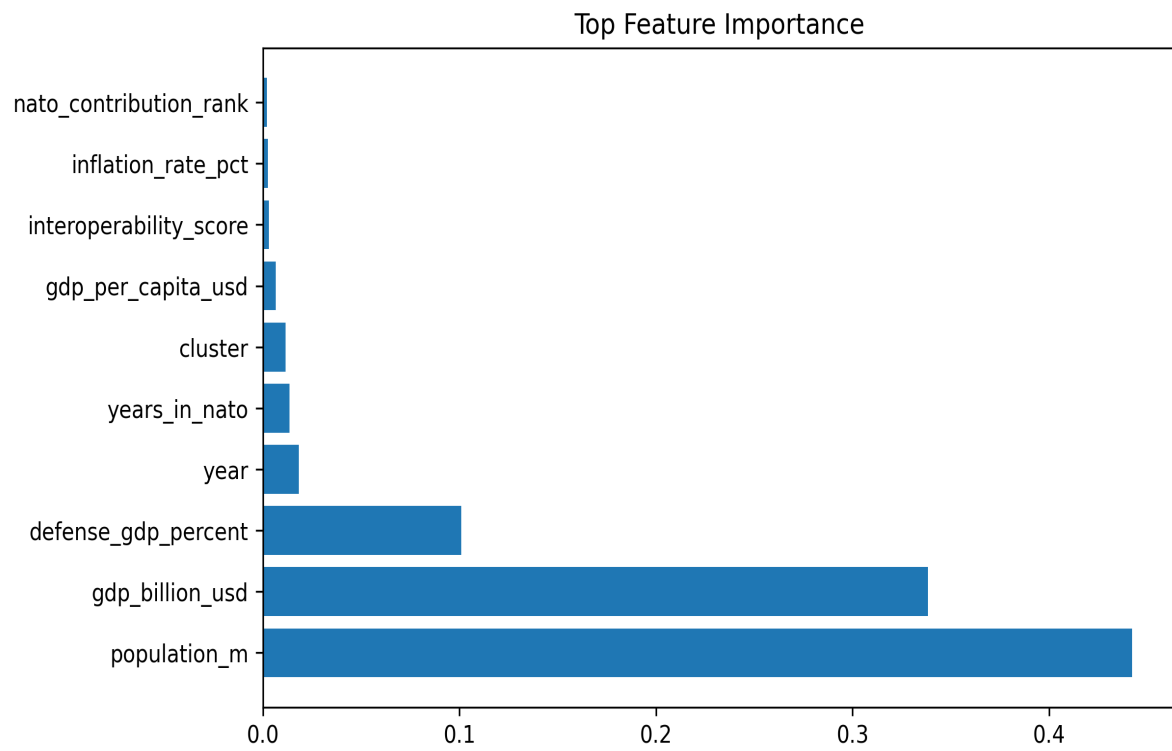
Timeline of NATO membership growth.



Validated KMeans clustering using silhouette optimization (k=2).



Isolation Forest anomaly detection identifying unusual observations.



Random Forest feature importance.

Final Dataset Summary

Processed Datasets:

- NATO_2_Equipment_Inventory.csv

Rows: 4056 | Columns: 26 | Target: combat_ready_pct | Charts: 17

- NATO_3_Operations_Missions.csv

Rows: 5200 | Columns: 31 | Target: mission_cost_m_usd | Charts: 16

- NATO_1_Country_Stats.csv

Rows: 1444 | Columns: 28 | Target: defense_budget_billion_usd | Charts: 18